Mini Project Report On

AceAIre - AI POWERED PLACEMENT ASSISTANT

Submitted in partial fulfillment of the requirements for the award of the degree of

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in

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CERTIFICATE

This is to certify that the mini project report entitled **"AceAIre—AI POWERED PLACEMENT ASSISTANT"** is a bonafide record of the work done by **Adithya Narayanan K P(U2303016)**, **Angel Vivek K(U2303041)**, **Anna Dantis (U2303047),** and **Avin Varghese (U2303064)**, submitted to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2025-2026.

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Abstract

The AI-Powered Placement Preparation Assistant is an intelligent platform designed to help engineering students prepare effectively for campus placements. The system provides a comprehensive preparation environment that includes technical practice, aptitude tests, group discussion practice, and mock interview sessions. The platform integrates Artificial Intelligence technologies such as speech-to-text processing, natural language processing, and computer vision to analyze student responses and provide meaningful feedback. Voice responses from users are converted to text using speech recognition models such as Whisper, and responses are evaluated using Large Language Models (LLMs) such as LLaMA through Ollama. Additionally, the system uses MediaPipe and OpenCV to analyze user behavior during mock interviews by evaluating eye contact, posture, and facial expressions. The application also provides performance analytics, progress tracking, weekly reports, and technology news updates to help students stay informed about industry trends. The system aims to bridge the gap between academic learning and industry expectations by providing a personalized and interactive placement preparation platform that helps students identify weak areas and improve their skills.

Keywords: Artificial Intelligence, Placement Preparation, Speech Recognition, Large Language Models, Natural Language Processing.

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List of Abbreviations

- AI - Artificial Intelligence
- LLM - Large Language Model
- NLP - Natural Language Processing
- SRS - Software Requirements Specification
- API - Application Programming Interface
- TTS - Text-to-Speech
- GD - Group Discussion
- REST - Representational State Transfer
- ORM - Object Relational Mapping
- CSE/IT - Computer Science Engineering / Information Technology

Chapter 1

Introduction

1.1 Background

Campus placement preparation is an important stage in the academic life of engineering students. To succeed in placement interviews, students need strong technical knowledge, logical thinking ability, and good communication skills. However, many students do not have access to proper platforms that guide them and help them prepare effectively for placements. Traditional preparation methods mainly depend on study materials and manual practice. These methods do not provide instant feedback or proper analysis of a student's performance. Students also do not get enough guidance on communication skills, confidence, and how to behave during interviews. With the development of technologies such as speech recognition and machine learning, it is now possible to build systems that can automatically evaluate student responses. These technologies can help create interactive platforms that simulate real interview situations and give useful feedback. The AI-Powered Placement Preparation Assistant is designed to provide a complete platform that supports technical practice, aptitude training, group discussion practice, and mock interview evaluation using Artificial Intelligence technologies.

1.2 Problem Definition

Students currently lack a single platform that provides comprehensive placement preparation including technical practice, aptitude training, group discussion evaluation, and mock interview simulation. The objective of this project is to develop an AI-powered placement assistant that provides structured practice and intelligent feedback to improve placement readiness.

1.3 Scope and Motivation

The proposed system provides a platform that helps students prepare for campus placements in a structured way. It allows students to practice technical questions and aptitude tests to improve their problem solving skills. The platform also includes simulated group discussions and mock interview sessions to help students gain real interview experience. The system evaluates student responses using AI-based technologies and provides feedback on their performance. This helps students understand their strengths and weaknesses and improve their preparation. The motivation for this project comes from the increasing competition in campus placements. Many students face difficulties during interviews because they do not get enough opportunities to practice in a real interview environment. Traditional preparation methods mainly rely on study materials and do not provide proper feedback. A smart system that can evaluate responses and give useful suggestions can help students improve their skills. Therefore, this project aims to use Artificial Intelligence to support students in preparing more effectively for placements.

1.4 Objectives

1. To develop a platform that helps students prepare for campus placements.
2. To provide technical and aptitude practice modules based on different branches.
3. To create a system that evaluates group discussion performance.
4. To develop a mock interview system that uses a camera to observe and analyze responses.
5. To analyze student performance and identify weak areas.
6. To generate progress reports and provide suggestions for improvement.

1.5 Challenges

- Integrating multiple AI technologies such as speech recognition, natural language processing, and computer vision into a single system is challenging.

- Efficiently processing voice and video data in real time also requires optimized resource management.

1.6 Assumptions

1. Users have access to a microphone and webcam.
2. The system has access to a stable internet connection.
3. The datasets used contain sufficient technical and aptitude questions.

1.7 Societal / Industrial Relevance

The project helps students improve employability skills and prepare effectively for job interviews. Educational institutions can also use such systems to analyze student performance and provide better placement training.

1.8 Organization of the Report

This report is organized into several chapters:

- Chapter 1 presents the introduction, including the background, objectives, scope, and motivation of the project.
- Chapter 2 describes the Software Requirements Specification (SRS), which includes the functional and non-functional requirements of the system.
- Chapter 3 explains the system architecture and design.
- Chapter 4 discusses the results obtained from the system and provides an analysis of the performance and functionality.
- Chapter 5 presents the conclusion of the project and summarizes the overall outcomes.
- Chapter 6 lists the bibliography, which includes the references used during the project development.
- Chapter 7 contains the appendix.

Chapter 2

Software Requirements Specification

2.1 Introduction

The Introduction section of an SRS explains the overall purpose and scope of the system being developed. In this project, the system is the AI Placement Assistant, which is designed to help students prepare for job placements using artificial intelligence. Many students face difficulties during placement interviews because they lack practice in technical questions, communication skills and confidence. This platform aims to solve that problem by creating a realistic environment where students can practice interviews and receive automated feedback.

The system provides several important features that simulate the real recruitment process. Students can participate in mock interviews, where they answer questions while their video and audio are recorded. The system then evaluates their performance using AI based analysis. It also includes group discussion simulations, where students practice speaking and presenting their ideas in a discussion setting. Additionally, the system provides news updates and industry information, helping students stay aware of current technological trends and job opportunities.

Another important part of the system is the Teacher Module. This module allows faculty members to monitor the progress of students and analyze their interview performance. Teachers can see metrics such as confidence score, eye contact level and overall interview score. They can also compare student performance across different academic branches.

By combining AI analysis with an interactive learning platform, the AI Placement Assistant aims to improve students' communication skills, technical readiness and overall confidence during real placement interviews.

2.2 Overall Description

The overall description section provides a high level overview of how the system works, who uses it and the environment in which it operates.

The AI Placement Assistant is designed as a client server architecture system. The frontend of the application is developed using Flutter, which allows the application to run on multiple platforms such as Android, web browsers, and potentially iOS devices. Flutter provides a responsive user interface and ensures that the same codebase can support different devices.

The backend of the system is implemented using Python, along with FastAPI. FastAPI is a modern backend framework designed to handle asynchronous API requests efficiently. It is responsible for managing user authentication, storing interview results, processing video and audio data and running machine learning models for behavioral evaluation.The system primarily serves two types of users:

- Students are the main users of the platform. They use the system to practice mock interviews, participate in group discussions, check their rankings among peers and track their improvement over time.

- Teachers use the system to analyze student performance. They can view interview reports, compare results between students and monitor the overall performance of students in their department or branch.

The system also interacts with hardware components such as webcams and microphones. These devices are required for recording video and audio during mock interviews. The captured media data is analyzed by AI algorithms to detect behavioral factors like eye contact, facial expressions and speech clarity.

Another important consideration in this system is data privacy and security. Since the platform collects sensitive data such as video recordings and personal performance reports, strict security mechanisms must be implemented. These include secure authentication, encrypted data storage and restricted access controls to ensure that only authorized users can view the data.

2.3 External Interface Requirements

The External Interface Requirements section describes how the system interacts with users, hardware devices and other software systems.

- User Interface

  The user interface is designed to be simple, modern and easy to navigate. Students are presented with a dashboard that provides quick access to various features such as starting a mock interview, joining a group discussion session, checking performance scores and reading industry news.

  The student dashboard also includes branch rankings, which allow students to see their position relative to other students in the same academic branch. This creates a competitive learning environment and motivates students to improve their performance.

  The teacher dashboard is more analytical in nature. Teachers can view student performance reports, access interview history and analyze trends in student progress. The interface also provides filters that allow teachers to view data branch wise or department wise.

- Hardware Interface

  The system interacts with two main hardware components:

  - Camera (Webcam) : The camera is used to record the student's video during a mock interview. This video feed is analyzed by AI algorithms to evaluate eye contact and engagement.
  - Microphone : The microphone captures the student's speech. The audio data is analyzed to evaluate speech clarity, confidence level and communication skills.

  The application requests permission from the user before accessing these devices.

- Software Interface

  The system relies on several software interfaces to function effectively.

The frontend application communicates with the backend server using RESTful APIs. These APIs send requests and responses between the user interface and the server.

The backend also communicates with a database using SQLAlchemy, which helps manage and store structured data such as user accounts, interview scores, and session history.

2.4 System Features

The system features section describes the main functional modules of the platform.

- Mock Interview Module:

  This is the core feature of the AI Placement Assistant. In this module, students participate in simulated interviews. The system presents interview questions that may include technical questions, HR questions or general aptitude questions.

  During the interview session, the student's video and audio are recorded. The backend system processes this data in real time using AI algorithms. These algorithms analyze several behavioral aspects, such as:

  - Eye contact with the camera
  - Confidence level
  - Voice

  After the interview is completed, the system generates a detailed report showing the student's performance and areas for improvement.

- Group Discussion Module:

  The Group Discussion module simulates a real GD environment. Students join a discussion room where they can speak about a given topic for a fixed duration.

  Before entering the discussion session, students are shown instructions explaining the rules of participation. The system ensures that discussions are structured and time bound.

  This module helps students improve their communication skills and ability to express ideas clearly.

- Teacher Module:

  The Teacher Module acts as a monitoring and management system for educators. Teachers can view individual student reports, compare scores and track improvement trends.

  Teachers can also analyze the performance of entire branches or departments. This helps them identify students who need additional support or training.

- News Module

  The News Module provides curated industry updates and technological news. These updates help students stay informed about current trends in the job market, new technologies and upcoming placement drives.

  This feature ensures that students not only improve their interview skills but also remain aware of industry developments.

2.5 Other Nonfunctional Requirements

Nonfunctional requirements define the quality attributes of the system rather than specific functionalities.

- Performance: The system must process video frames and audio signals quickly to provide real time feedback. AI algorithms should analyze camera metrics without noticeable delays so that students receive instant results after completing an interview.

- Security: Security is a critical aspect of the system. The platform must ensure that user data is protected from unauthorized access. Authentication mechanisms should verify user identity before granting access to the system. Additionally, all stored data must be encrypted to prevent data breaches.

- Reliability: The system must be reliable even during high usage periods, such as during university placement seasons when many students may be using the platform simultaneously. The backend should be capable of handling multiple interview sessions without crashing or slowing down.

- Usability: The application must provide a smooth and engaging user experience. The interface should be intuitive so that students and teachers can easily navigate the system. Features such as loading animations, clear navigation menus and organized dashboards improve usability and make the platform more professional.

Chapter 3

System Architecture and Design

3.1 System Overview

The AI-Powered Placement Preparation Assistant is designed to support students in preparing for campus placements by providing an integrated platform for technical practice, aptitude tests, group discussions, and mock interviews. The system combines modern technologies such as artificial intelligence, speech recognition, and computer vision to evaluate user responses and provide meaningful feedback. The objective of the system is to create an environment where students can practice placement-related activities and improve their skills before attending real recruitment processes.

3.2 Architectural Design

The architecture of the system follows a client–server model with a layered design, which separates the user interface, application logic, AI processing, and data storage components. This architecture improves scalability, maintainability, and efficient communication between different modules. The major components of the system include the User Layer, Flutter Desktop Application, Backend Server, AI Processing Layer, Database Layer, and External APIs, as illustrated in the system architecture diagram.

At the top level is the User, which includes both students and teachers. Students primarily use the platform to practice tests, participate in group discussion sessions, and attend mock interview simulations. Teachers or administrators can view performance reports and analyze the progress of students. The interaction with the system begins when the user registers and logs into the application.

The Flutter Desktop Application acts as the frontend interface of the system. It provides an interactive dashboard where users can access different preparation modules such as aptitude tests, technical tests, group discussions, and mock interviews. The mobile

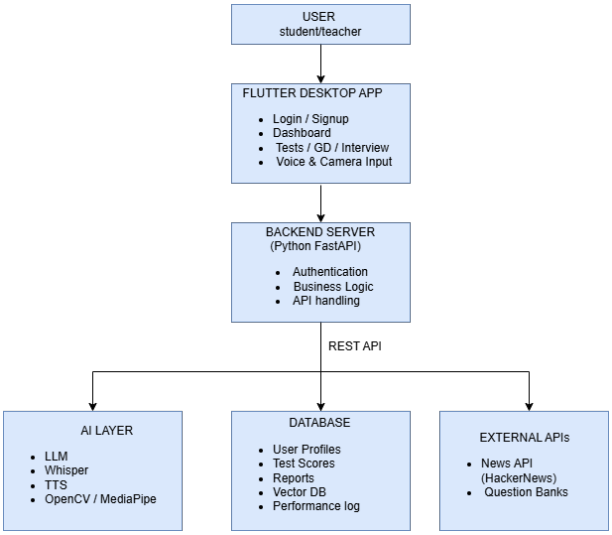

Figure 3.1: Architectural Design

application also captures voice input through the microphone and video input through the device camera during GD and interview sessions. These inputs are sent to the backend server for further processing and evaluation. Communication between the frontend and backend is handled using REST API calls.

The Backend Server, developed using Python FastAPI, acts as the central controller of the application. It handles authentication, manages API requests, and executes the business logic of the system. When a user performs an action such as starting a test or submitting an answer, the request is sent to the backend server. The server processes the request, retrieves relevant information from the database, and communicates with the AI layer when evaluation or analysis is required.

The AI Processing Layer performs intelligent analysis of user responses. This layer includes technologies such as Large Language Models (LLMs), Whisper speech-to-text processing, Text-to-Speech (TTS), and computer vision frameworks such as OpenCV and MediaPipe. The Whisper model converts spoken responses into text during group discussions and interviews. The text is then analyzed using the language model to evaluate the quality, relevance, and clarity of the response. Computer vision techniques can also analyze facial expressions and eye contact to estimate the user's confidence and engagement.

The Database Layer stores all important data required for the system. This includes user profiles, test questions, answers submitted by users, test scores, transcripts, and generated reports. Performance logs are also maintained in the database to track the

progress of students over time. Efficient data storage ensures that the system can quickly retrieve and update information when required.

In addition to internal modules, the system also interacts with External APIs to enhance functionality. For example, technical news and updates can be retrieved from sources such as the Hacker News API, which helps students stay informed about current industry trends. External question banks may also be integrated to increase the number of practice questions available to users.

Overall, the system operates by allowing the user to interact with the Flutter application, which sends requests to the backend server. The backend processes the request, communicates with the AI layer and database, and retrieves additional information from external APIs when needed. The processed results, including scores, feedback, and reports, are then returned to the frontend and displayed to the user. This modular architecture ensures efficient operation and allows the system to be extended with additional features in future versions.

3.3 Dataset Identified

The proposed system uses multiple datasets related to technical questions, aptitude questions, group discussion topics, and interview preparation. These datasets are collected from publicly available educational platforms and online repositories. The datasets contain questions, answers, that are used in different modules of the system such as technical practice, aptitude tests, group discussion practice, and interview preparation.

Technical Question Dataset

- CSE / IT / CSBS / AIDS:
  https://www.kaggle.com/datasets/keithzidandsouza/engineering-aptitude-test-questions?select=
- ECE / AEI:
  https://engineeringinterviewquestions.com/electronics-communication-engineering-questions-answers/
- Civil Engineering:
  https://engineeringinterviewquestions.com/civil-engineering-multiple-choice-interview-questions-and-answers/

- Mechanical Engineering:
  https://engineeringinterviewquestions.com/mechanical-engineering-multiple-choice-questions-and-answers/

- Electrical and Electronics Engineering (EEE):
  https://engineeringinterviewquestions.com/electrical-engineering-multiple-choice-interview-questions-and-answers/

Aptitude Dataset

- https://www.kaggle.com/datasets/keithzidandsouza/engineering-aptitude-test-questions/code

Group Discussion Topics Dataset

- https://www.indiabix.com/group-discussion/topics-with-answers/

Interview Topics Dataset

- CSE / IT / AIDS:
  https://engineeringinterviewquestions.com/computer-science-engineering-interview-questions-and-answers-pdf-cse/

- ECE / AEI:
  https://engineeringinterviewquestions.com/electronics-communication-engineering-questions-answers/

- Civil Engineering:
  https://engineeringinterviewquestions.com/civil-engineering-multiple-choice-interview-questions-and-answers/

- Mechanical Engineering:
  https://engineeringinterviewquestions.com/mechanical-engineering-multiple-choice-questions-and-answers/

- Electrical and Electronics Engineering (EEE):
  https://engineeringinterviewquestions.com/electrical-engineering-interview-questions-and-answers-eee/

3.4 Proposed Methodology/Algorithms

This section describes the methodologies used in the AI Powered Placement Assistant system. The system combines adaptive learning techniques and AI based evaluation to provide an effective placement preparation platform. Different modules such as aptitude tests, technical practice, group discussion simulation, and interview practice collect user responses and analyze them using automated evaluation mechanisms.

The methodology focuses on monitoring user performance, generating feedback, and dynamically adjusting the difficulty level of questions. Based on the user's past performance and accuracy, the system selects appropriate questions and gradually increases or decreases the difficulty level. This approach ensures that students receive personalized practice while improving their technical knowledge, communication ability, and interview confidence.

3.4.1 Performance Evaluation Algorithm

Purpose: To evaluate the student's responses in aptitude and technical tests and calculate the overall performance score.

Algorithm Steps:

1. Receive the username, category (Aptitude/Technical), list of question IDs, and user answers.
2. Initialize variables such as *correct_count* and *total_questions*.
3. For each question, retrieve the correct answer from the question database.
4. Compare the user answer with the correct answer.
5. If the answer is correct, increment the correct count.
6. Store each question response and correctness information in the database.
7. After evaluating all responses, compute the final score using:

$$Score = \frac{Correct\ Answers}{Total\ Questions} \times 100$$

8. Store the final score and test details in the results table.

9. Return the score and feedback to the user dashboard.

3.4.2 Adaptive Difficulty Adjustment Algorithm

Purpose: To automatically adjust the difficulty level of questions based on the user's performance over time.

Algorithm Steps:

1. Retrieve the user's current difficulty level and previous test results from the database.

2. Calculate the average performance percentage from recent tests.

3. Count the number of days the user completed practice sessions successfully.

4. If the user consistently scores above a predefined threshold (for example 75%), increase the difficulty level.

5. If the user performs at an average level, maintain the current difficulty level.

6. If the performance is below the threshold, reduce the difficulty level to support further practice.

7. Update the user's difficulty level in the user database.

8. Store the weekly performance summary for future analysis.

9. Continue monitoring the performance for further adjustments.

3.5 User Interface Design

The user interface design (wireframe designs) can be highlighted in this section. The figures titles should be in a chronological order and self explanatory.

3.6 Database Design

The database is an important component of the AI Powered Placement Assistant system as it stores user information, question banks, responses, and performance data generated

during placement preparation activities. The system uses a relational database approach to ensure structured storage, efficient data retrieval, and reliable management of user records.

The project uses MySQL as the primary database management system. The database is accessed through Python libraries such as PyMySQL and SQLAlchemy ORM. SQLAlchemy provides an object relational mapping mechanism that allows database tables to be represented as Python classes. This makes database operations easier to implement and improves code maintainability.

MySQL was selected for this system because it provides several advantages suitable for the project requirements. First, it supports relational integrity, which ensures that data stored in different tables such as users, quiz results, and evaluation records remain properly linked using keys and constraints. This prevents data inconsistencies and maintains structured relationships between records. Another important advantage of MySQL is its ability to handle multiple users simultaneously. Since the placement assistant platform may be accessed by several students at the same time, the database must support concurrent read and write operations. MySQL, using the InnoDB storage engine, supports row-level locking and ensures reliable transaction management. MySQL also provides efficient indexing and query performance, which is useful when retrieving large amounts of data such as test results, analytics records, and historical performance information. These capabilities make MySQL a suitable choice for the system.

The database schema is organized into three major categories: User Management, Content Storage, and Performance Tracking. The users table stores information related to authentication and user profiles. It includes attributes such as user ID, username, email, password hash, engineering branch, role (student or teacher), and difficulty levels for aptitude and technical modules. These attributes help the system personalize the learning experience for each user. The questions table stores aptitude and technical multiple-choice questions used in the practice modules. It includes the question text, answer options, correct answer, category, difficulty level, branch information, and explanation for the answer. This table forms the main question bank used during practice sessions. Additional tables such as interview questions and group discussion topics store questions used for mock interviews and group discussion sessions. These tables contain text prompts and reference answers used by the AI evaluation module.

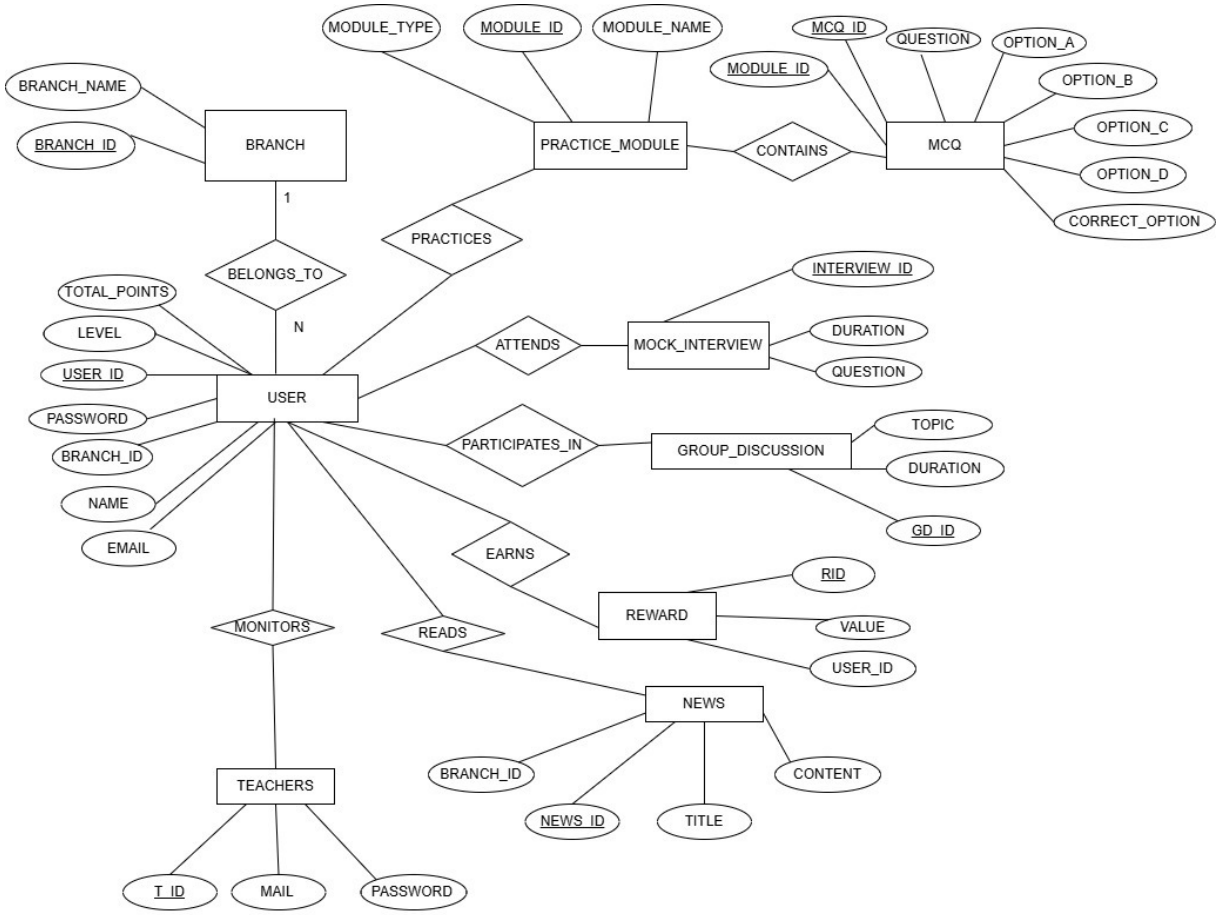

Figure 3.2: ER DIAGRAM

To track user performance, the system uses several tables such as results, quiz answers, and group discussion evaluations. The results table stores summary information such as score, total questions, module category, and timestamp. The quiz answers table stores detailed response data for each question attempted by the user.

For GD and interview modules, the gd evaluations table stores the transcript generated using speech to text processing along with scores related to communication, content quality, and behavioral analysis. The system also maintains a weekly statistics table that records user activity levels, average scores, and progress over time.

3.7 Description of Implementation Strategies

This section describes the important implementation strategies used in the AceAIre system. The application integrates computer vision, speech recognition, and natural language processing techniques to evaluate student performance during mock interviews and group

discussions. Python-based libraries are used for video processing, audio analysis, and AI-based evaluation.

3.7.1 Video Processing and Analysis

To analyze the student's behavior during mock interviews and group discussions, the system processes the recorded video submitted by the frontend application. The Flutter client records the video and sends the `.mp4` file to the Python backend through REST APIs. The backend then extracts frames and analyzes facial behavior.

Libraries Used

- OpenCV (`cv2`) – video frame extraction and processing
- MediaPipe – facial landmark detection
- NumPy – numerical computation

Implementation Strategy

1. The recorded video file submitted by the user is processed using the `cv2.VideoCapture()` method.
2. Frames are extracted sequentially from the video file.
3. Each frame is converted from BGR to RGB format before processing.
4. MediaPipe Face Mesh detects facial landmarks on the user's face.
5. Selected landmark points are used to estimate head orientation and determine whether the user maintains eye contact with the camera.

Code Snippet

```
import cv2
cap = cv2.VideoCapture(video_path)
while cap.isOpened():
ret, frame = cap.read()
if not ret:
break
```

```
frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
cap.release()
```

3.7.2 Head Pose Estimation

Head pose estimation is used to determine the orientation of the student's head during the interview. By analyzing the positions of facial landmarks such as the nose tip, chin, eye corners, and mouth corners, the system calculates the Pitch, Yaw, and Roll angles. This is implemented using OpenCV's `solvePnP()` algorithm.

Code Snippet

```
def _get_pose(landmarks, img_w, img_h):
image_points = np.array([
(landmarks[1].x * img_w, landmarks[1].y * img_h),
(landmarks[152].x * img_w, landmarks[152].y * img_h),
(landmarks[33].x * img_w, landmarks[33].y * img_h),
(landmarks[263].x * img_w, landmarks[263].y * img_h),
(landmarks[61].x * img_w, landmarks[61].y * img_h),
(landmarks[291].x * img_w, landmarks[291].y * img_h)
], dtype=np.float64)
focal_length = img_w
center = (img_w / 2, img_h / 2)
camera_matrix = np.array([
    [focal_length, 0, center[0]],
    [0, focal_length, center[1]],
    [0, 0, 1]
], dtype=np.float64)
dist_coeffs = np.zeros((4, 1))
success, rotation_vector, translation_vector = cv2.solvePnP(
    self.model_points, image_points, camera_matrix, dist_coeffs,
    flags=cv2.SOLVEPNP_ITERATIVE
)
rmat, _ = cv2.Rodrigues(rotation_vector)
```

```
angles, _, _, _, _, _ = cv2.RQDecomp3x3(rmat)

return angles[0], angles[1], angles[2]  # Pitch, Yaw, Roll
```

These angles are used to determine whether the student is looking directly at the camera or frequently looking away.

3.7.3 Audio Processing and Speech Recognition

The system analyzes the spoken responses of the student by extracting audio from the recorded session and converting it into text. **Libraries Used**

- Librosa – audio signal processing and silence detection
- Whisper – speech-to-text transcription

Implementation Strategy

1. The recorded audio is first analyzed using `librosa.effects.split()` with a targeted decibel threshold (`top`$_db\approx$ `25`) to isolate segments containing actual speech and filter out background noise.
2. If sufficient speech activity is detected, the audio is passed to the Whisper model.
3. Whisper converts the spoken English audio into an accurate text transcript.

Code Snippet

```
import whisper
model = whisper.load_model("base")
def transcribe_audio(audio_file):
result = model.transcribe(audio_file)
return result["text"]
```

3.7.4 AI-Based Response Evaluation

After the transcript is generated, the system evaluates the student's response using a Large Language Model (LLaMA) running locally through the Ollama framework. **Libraries Used**

- Ollama Python API
- JSON parsing utilities

Implementation Strategy The transcript and interview topic are combined into a structured prompt. The LLM analyzes the response and produces evaluation results in JSON format, which can be easily parsed and stored in the database. The model configuration is optimized for stable outputs:

- `temperature = 0.1` to reduce randomness
- JSON formatted responses for structured evaluation

Code Snippet

```
import ollama
def evaluate_response(prompt):
try:
response = ollama.generate(
model="llama3.2:1b",
prompt=prompt,
format="json",
options={
"temperature": 0.1,
"num_predict": 500
}
)
    return response['response']
except Exception as e:
    print(f"Ollama Library Error: {e}")
    return "{}"
```

3.7.5 Performance Evaluation Metrics

The final evaluation of a student's performance is computed using multiple metrics derived from the video, audio, and language analysis modules. The metrics include:

- Content Score : evaluates the correctness and relevance of the response
- Communication Score : evaluates clarity and structure of speech

$$comm_score = min(10, (LLM_comm_sc * 0.6) + (v_sc * 0.4) + in_sc) \quad (3.1)$$

- Camera Score : evaluates eye contact, posture, and visibility

$$camera_score = round(max(0, min(10, raw_score + penalties)), 1) \quad (3.2)$$

- Voice Score: measures speaking frequency and activity level

$$voice_score = (wpm_score * 0.4) + (silence_score * 0.3) + (filler_score * 0.3) \quad (3.3)$$

- Final Aggregated Score : combined score calculated from all evaluation components

$$final_score = round(cont_score * 0.5 + comm_score * 0.3 + cam_score * 0.2) \quad (3.4)$$

These evaluation results are stored in the MySQL database and later visualized in the student analytics dashboard.

3.8 Module Division

The proposed system is divided into several modules to ensure efficient functionality, modular development, and easy maintenance. Each module performs a specific task within the system and interacts with other modules through the backend server. The following are the major modules included in the AI-Powered Placement Preparation Assistant system.

1. User Module

The User Module provides the common registration and login functionality for users. It allows users to create accounts and select their role as either a student or teacher. Students can access practice modules, view reports, and monitor their progress through the dashboard. Teachers can view student performance analytics and reports. This module securely stores user profiles, authentication details, and role information in the database.

2. Technical Test Module

The Technical Test Module manages branch-specific technical questions stored in the database. The backend server retrieves questions and sends them to the frontend through REST APIs. Students attempt the tests, and their responses are submitted to the backend for evaluation. The system calculates scores and adjusts the difficulty level based on the student's previous performance. All test results and feedback are stored for future analysis.

3. Aptitude Test Module

The Aptitude Test Module maintains a database of aptitude questions covering various topics such as quantitative aptitude, logical reasoning, and verbal ability. The module delivers multiple-choice tests through the frontend interface. After submission, the backend evaluates the answers and calculates scores automatically. The system stores results and uses them to analyze student performance and track progress over time.

4. Group Discussion (GD) Practice Module

The GD Practice Module allows students to practice group discussion topics stored in the database. During a GD session, the system captures the user's voice through the microphone. The recorded speech is converted into text using the Whisper speech-to-text model. The text is then analyzed by an AI model to evaluate communication quality and content relevance. The system stores the performance data and feedback for further analysis.

5. Interview Practice Module

The Interview Practice Module simulates technical interview sessions for students. Interview questions are retrieved from the database and presented to the user. The system records both voice and video responses during the interview session. The spoken responses are evaluated using AI models such as LLaMA or Groq APIs, while the video input is used to analyze confidence and engagement. The system generates feedback and stores interview scores in the database.

6. News and Notification Module

The News and Notification Module retrieves technical news and industry updates from external sources such as the Hacker News API and RSS feeds. The system filters the news based on the user's selected branch and displays relevant updates on the dashboard. It can also send notifications and convert news text into speech using a Text-to-Speech (TTS) system.

7. AI Evaluation and Feedback Module

This module performs the core evaluation process of the system. User responses from different modules are sent to a Large Language Model (LLM) for analysis. The system checks keyword relevance, content quality, and communication clarity. Based on this analysis, it generates scores and improvement suggestions. The feedback is then returned to the user and stored in the database.

8. Analytics and Report Module

The Analytics and Report Module collects performance data from all modules of the system. It analyzes the results to identify strengths and weak areas in the user's performance. The system generates weekly or periodic reports and displays visual summaries such as graphs and charts on the dashboard to help users track their improvement.

9. Camera Evaluation Module

The Camera Evaluation Module captures real-time video input from the user during interview or GD sessions. Using computer vision libraries such as MediaPipe, the system detects facial landmarks and analyzes behavioral features such as eye contact, head movement, and blink rate. These factors help estimate the user's confidence and engagement during the session. The results are stored for further evaluation and feedback.

10. Voice Input Module

The Voice Input Module manages voice interactions within the system. It converts speech input into text using Whisper speech-to-text technology. The converted text is then processed by the backend server for evaluation or command processing. The module can also generate spoken responses using Text-to-Speech (TTS) technology to improve accessibility and interaction.

3.9 Work Schedule - Gantt Chart

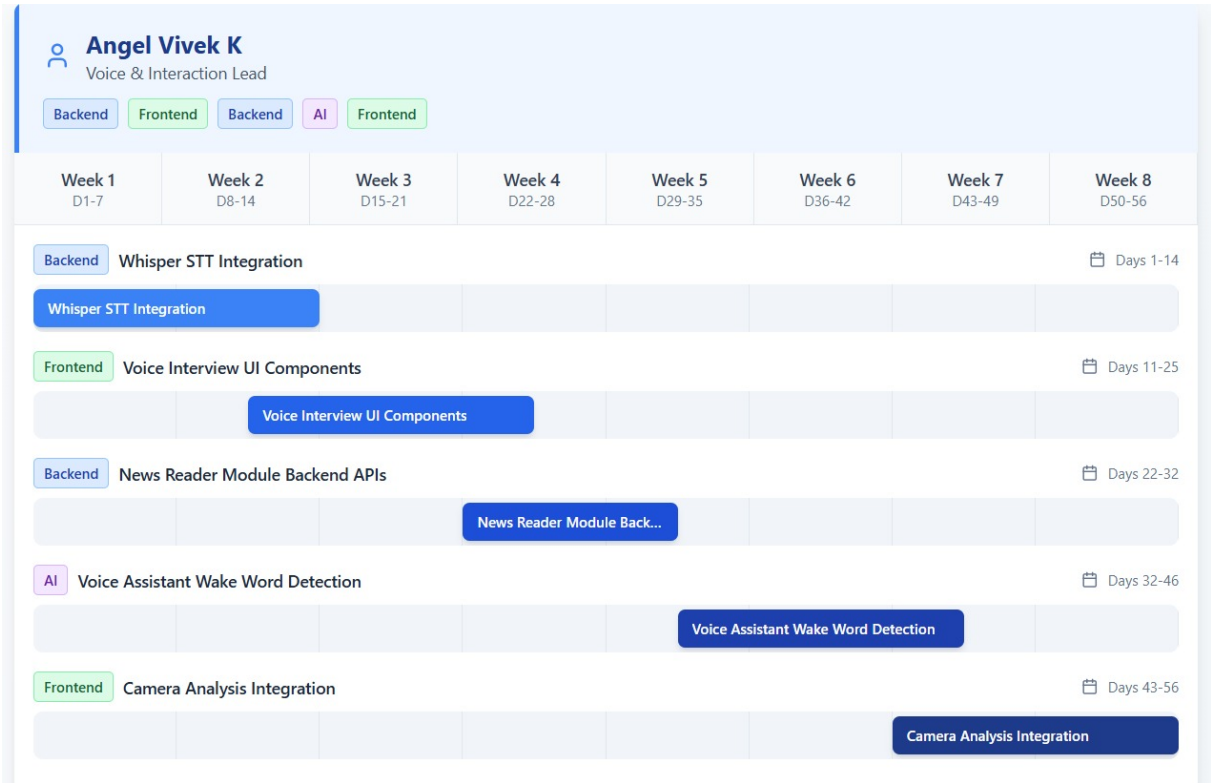

Figure 3.3: Gantt chart 1

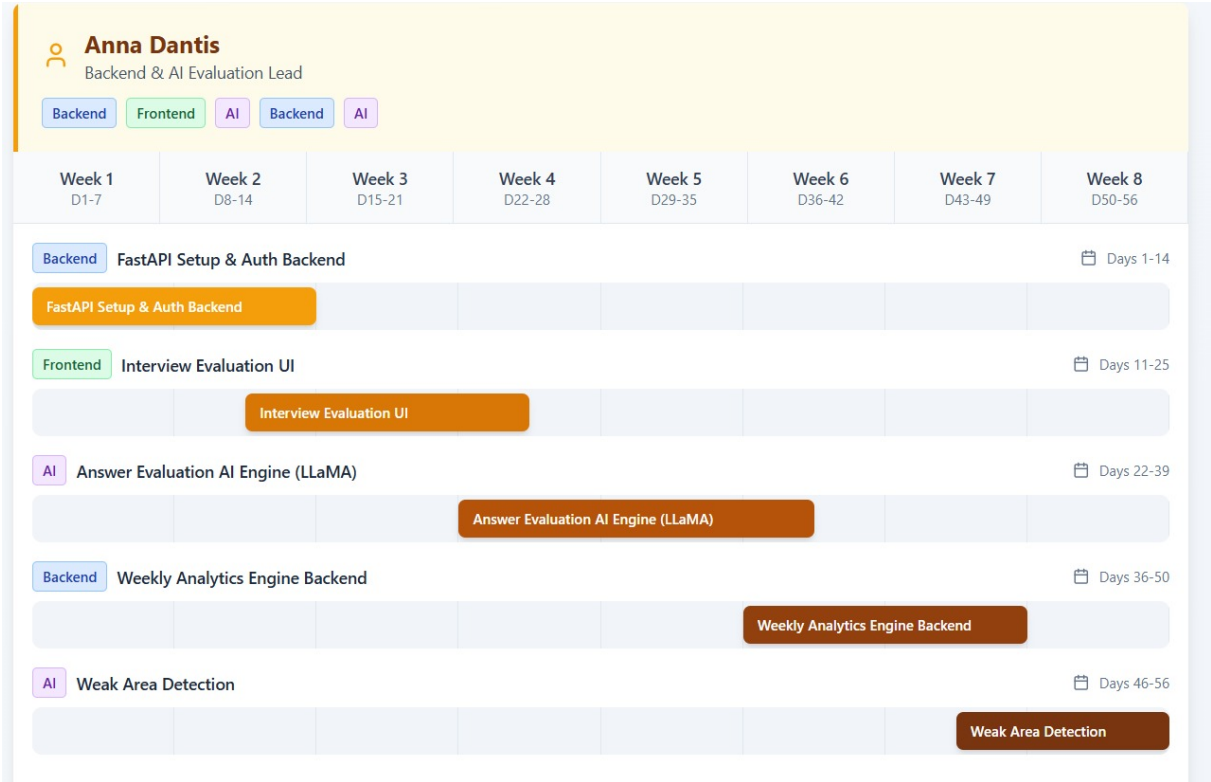

Figure 3.4: Gantt chart 2

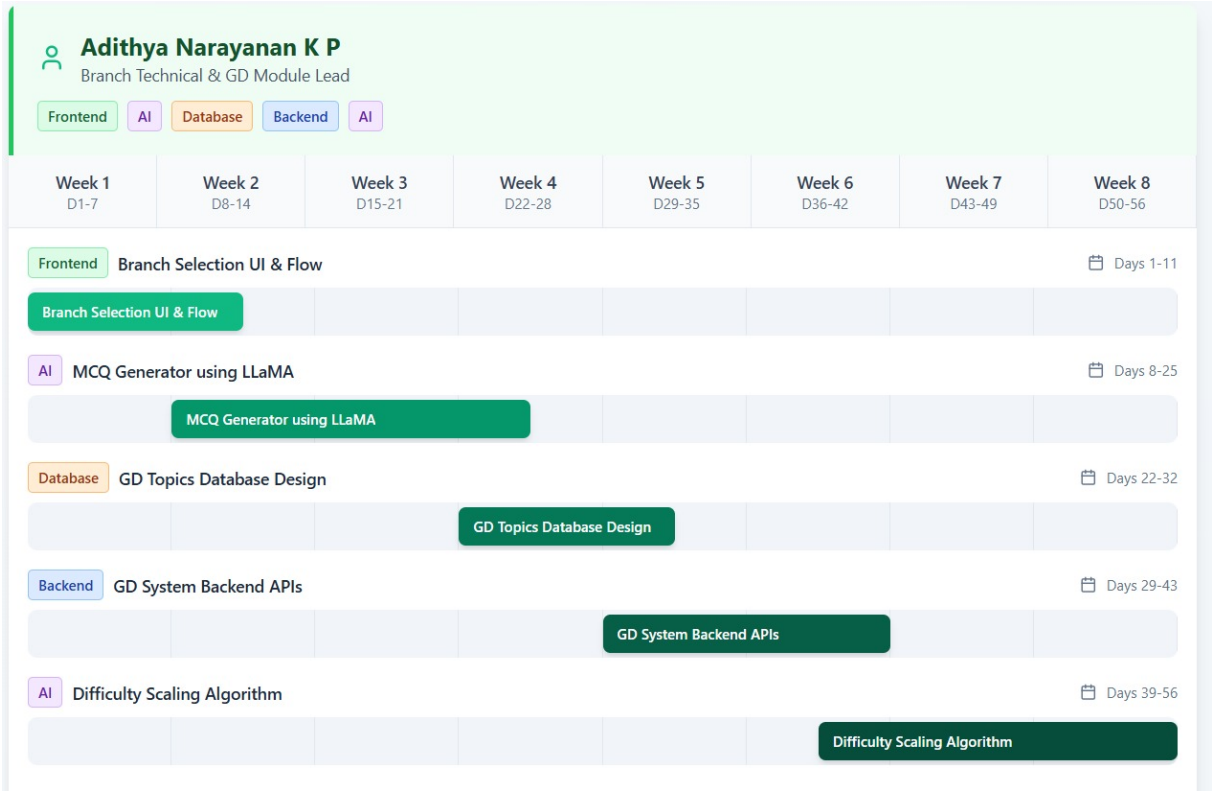

Figure 3.5: Gantt chart 3

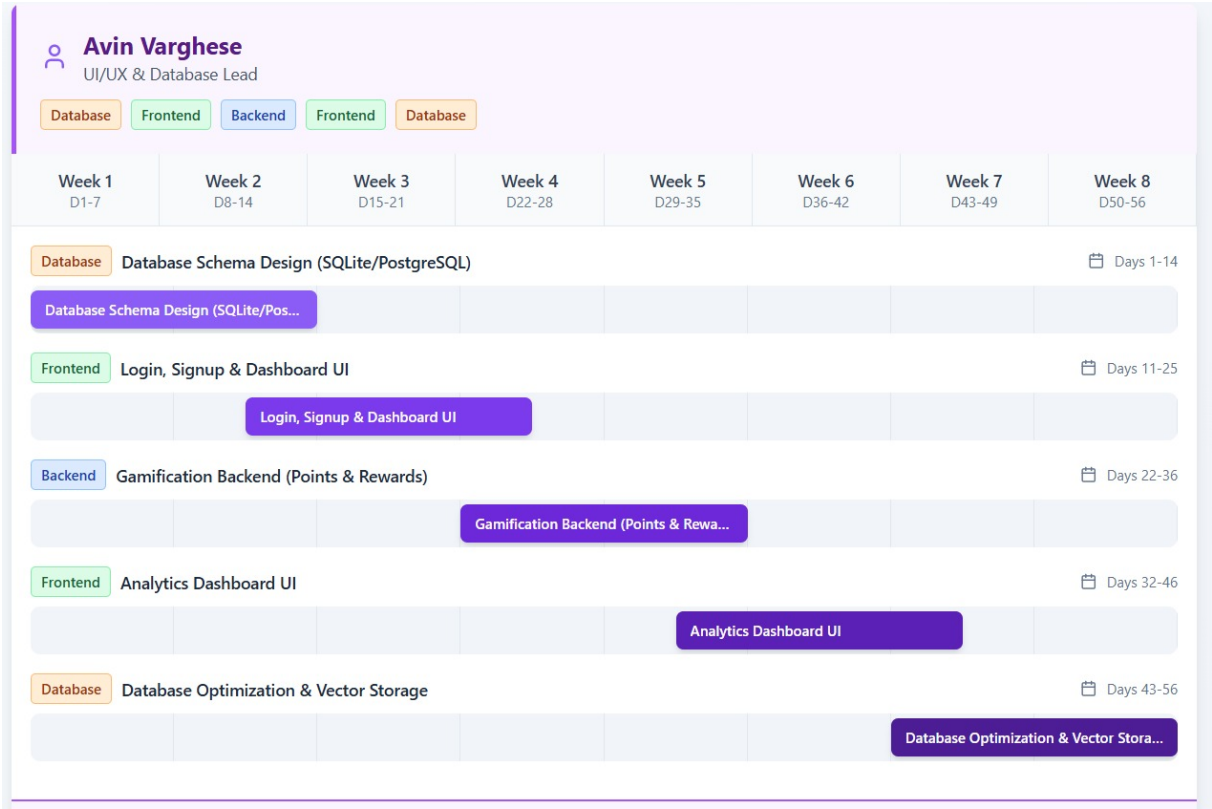

Figure 3.6: Gantt chart 4

Chapter 4

Results and Discussions

4.1 Overview

The AI Powered Placement Assistant system was successfully developed and tested across multiple modules including aptitude tests, technical tests, group discussions, and mock interviews. The system was able to process user inputs in the form of text, audio, and video and generate meaningful feedback using AI-based evaluation techniques. The integration of speech recognition, computer vision, and language models enabled the system to simulate real-world interview scenarios effectively.

4.2 Testing

The system was tested using different user scenarios to ensure proper functionality of each module.

- User Registration and Login functionality was tested for both students and teachers.
- Aptitude and Technical Test modules were tested with multiple datasets and response submissions.
- Mock Interview module was tested by recording video and audio inputs and verifying evaluation results.
- Group Discussion module was tested for speech-to-text conversion and response analysis.
- API endpoints were tested using Postman to ensure proper backend communication.

The outputs were verified by comparing expected results with actual results, and the system performed reliably across all test cases.

4.3 Discussion

The results demonstrate that the system effectively supports placement preparation through automated evaluation and feedback. The integration of AI technologies improved the accuracy of response evaluation and provided meaningful insights into student performance.

However, slight variations in speech recognition accuracy were observed due to background noise and accent differences. Similarly, camera-based evaluation depends on lighting conditions and camera quality. Despite these challenges, the system achieved satisfactory performance and fulfilled the project objectives.

Chapter 5

Conclusion

5.1 Conclusion

The AI Powered Placement Assistant system was successfully designed and implemented to support students in preparing for campus placements. The system integrates multiple technologies such as machine learning, speech recognition, and computer vision to create an interactive and intelligent learning environment.

The platform provides various features including aptitude tests, technical practice, group discussions, and mock interview simulations. It evaluates user performance and generates detailed feedback, helping students identify their strengths and weaknesses. The system bridges the gap between theoretical learning and practical interview preparation by simulating real-world scenarios.

Overall, the project demonstrates how Artificial Intelligence can be effectively used in the education domain to enhance learning experiences and improve employability skills.

5.2 Future Scope

The system can be further enhanced by introducing advanced features to improve functionality and user experience.

- **Peer-to-Peer Group Discussion:** The platform can be extended to allow multiple students to participate in real-time group discussions. The system can record conversations and evaluate each participant's communication skills, interaction level, and contribution.

- **Behavioral and Emotion Analysis:** Future versions can incorporate advanced computer vision techniques to analyze facial expressions, attention levels, and confidence during mock interviews for more accurate evaluation.

- **Expanded Question Bank and Coding Practice:** The system can include a larger collection of HR questions, soft skill questions, and coding problems to support preparation for various placement rounds, including technical interviews and coding assessments.

These enhancements can improve the scalability, intelligence, and overall effectiveness of the system, making it more beneficial for a wider range of users.

References

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[2] D. S. Prakash, S. KM, D. R, S. S. R and P. M. B, "An Integrated Web-Based Platform for Enhanced College Placement Management and Student Engagement," in 2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 2024, pp. 2624-2628, doi: 10.1109/ICACCS60874.2024.10717061.

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[4] A. Darain, S. Anjum, I. Khan, T. Shiekh and S. Ansari, "Placement Preparation Web-Application," International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE), vol. 12, 2023, doi: 10.17148/IJARCCE.2023.124158.

Appendix A: Presentation

AceAIre- AI POWERED PLACEMENT ASSISTANT

Mini Project Presentation

Team Members:
Adithya Narayanan K P – U2303016
Angel Vivek K – U2303041
Anna Dantis – U2303047
Avin Varghese – U2303064

Guide: Ms Reshma.M.R

Contents

- Introduction
- Problem Definition
- Objectives
- Scope and Relevance
- System Design
- Datasets
- Work Division – Gantt Chart
- Software/Hardware Requirements
- Results
- Conclusion
- Future Enhancements
- References

Introduction

- Campus placement preparation requires strong technical, aptitude, and communication skills.
- Most existing platforms lack personalized and branch specific training for students.
- Students do not receive real time feedback for GD and interview performance.
- The proposed AI powered placement assistant provides adaptive learning and evaluation.
- This application helps students identify weak areas and improve placement readiness efficiently.

Problem Definition

To design, develop, and implement an AI powered placement preparation system that provides personalized, branch specific technical training, aptitude practice, and AI based GD and interview evaluation on a single platform.

Objectives

- To develop a unified platform for placement preparation for all engineering branches.
- To provide branch specific technical tests and common aptitude practice.
- To conduct AI based group discussion and interview practice with real time feedback.
- To analyze user performance and identify weak areas using AI techniques.
- To generate weekly progress reports and recommendations for improvement.

Scope and Relevance

Scope

- AI-based quiz and GD practice platform
- Automatic speech to text conversion
- AI-based performance evaluation
- Student progress tracking system

Relevance

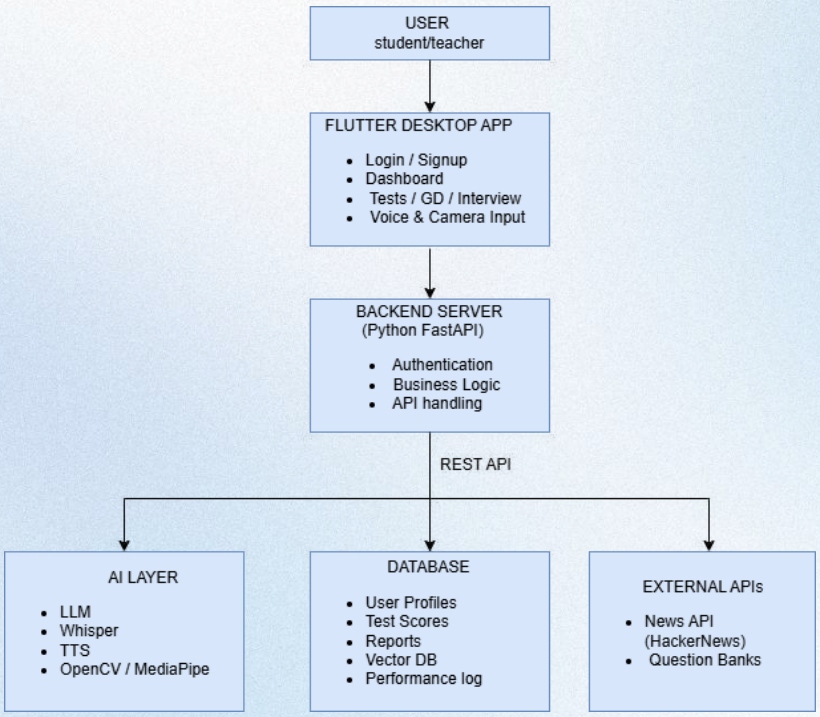
- Helps students prepare for campus placements
- Useful for college placement training programs
- Supports skill improvement in communication and technical areas

System Design

System Overview

- The system provides an AI based placement preparation platform for engineering students.
- Users register, select their branch, and access personalized learning modules.
- The application offers technical tests, aptitude practice, GD, and interview simulations.
- AI evaluates performance and generates feedback, reports, and recommendations
- Tracks daily performance and generates weekly analytical reports
- Includes voice assistant, notifications, and tech news updates
- Acts as a personalised virtual mentor for placement readiness

Architectural Design



Module Wise Explanation

1.USER MODULE

- Provides common registration and login with role selection (Student / Teacher)
- Student role accesses tests, practice modules, reports, and dashboard
- Teacher role accesses the student's performance and analytics.
- Stores user profiles, roles, and performance data securely

2.TECHNICAL TEST MODULE

- Store branch wise technical questions in database
- Send questions to frontend via REST API
- Accept user answers and send to backend
- Evaluate answers using AI model
- Adjust difficulty level based on past performance
- Store scores and feedback in database

3.APTITUDE TEST MODULE

- Maintain aptitude question bank in database
- Deliver MCQ tests through frontend
- Calculate scores automatically in backend
- Adjust difficulty level based on past performance
- Save test results for analytics

4.GD PRACTICE MODULE

- GD topics are stored in database
- Capture user voice during GD session
- Convert speech to text using Whisper.cpp
- Capture video feed from camera
- Evaluate the user's answer using AI
- Store GD performance data

5.INTERVIEW PRACTICE MODULE

- Technical interview questions are stored in database
- Record user voice and video responses
- Evaluate answers using LLaMA / Groq API
- Analyze confidence level using MediaPipe
- Store interview feedback and scores

6.NEWS & NOTIFICATION MODULE

- Fetch technical news using HackerNews API and RSS feeds
- Filter news based on user branch
- Display news on user dashboard
- Send daily notifications to users
- Convert news text to voice using TTS

7.AI EVALUATION & FEEDBACK MODULE

- Send user responses to LLM for analysis
- Perform keyword and content relevance checking
- Generate scores and improvement suggestions
- Return feedback to user
- Store AI-generated feedback in database

8.ANALYTICS & REPORT MODULE

- Collect performance data from all modules
- Analyze trends and weak areas using AI
- Generate weekly and yearly reports
- Create performance summaries and graphs
- Display analytics on dashboard

9.CAMERA EVALUATION MODULE

- Captures real time video input from the user
- Detects face and facial landmarks using MediaPipe
- Evaluates eye contact to measure confidence and engagement
- Checks attention by detecting continuous face presence on screen
- Measures blink rate to identify nervousness or stress
- Stores camera-based evaluation results for analysis and feedback

10. VOICE INPUT MODULE

- Convert speech to text using Whisper.cpp
- Process inputs in the backend
- Generate responses or actions
- Convert text to speech using TTS

11. CAMERA EVALUATION MODULE

- Captures real time video input from the user
- Detects face and facial landmarks using MediaPipe
- Evaluates eye contact to measure engagement
- Analyzes head tilt to assess posture and attentiveness
- Checks attention by detecting continuous face presence on screen
- Measures blink rate to identify nervousness or stress
- Stores camera based evaluation results for analysis and feedback

Algorithms

Performance Evaluation Algorithm

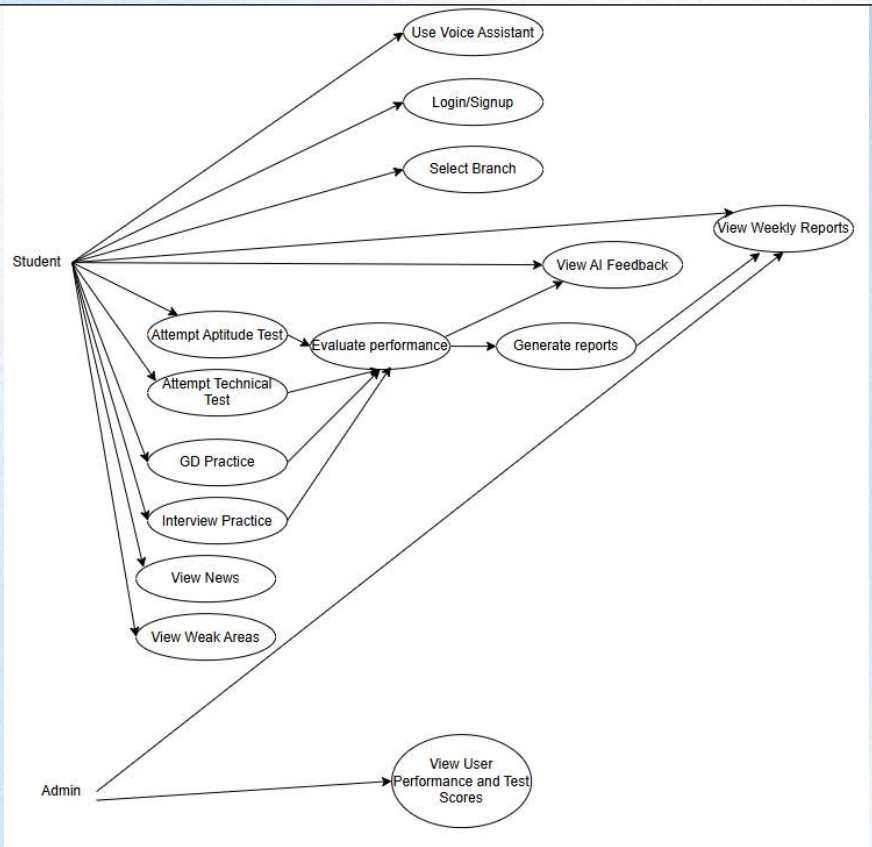
1. Start
2. Accept user response from the selected module
3. If input is voice, convert speech to text using STT
4. Extract keywords
5. Evaluate response using AI evaluation model
6. Generate score and feedback
7. Store results in database
8. Display feedback to user
9. Stop

Algorithms

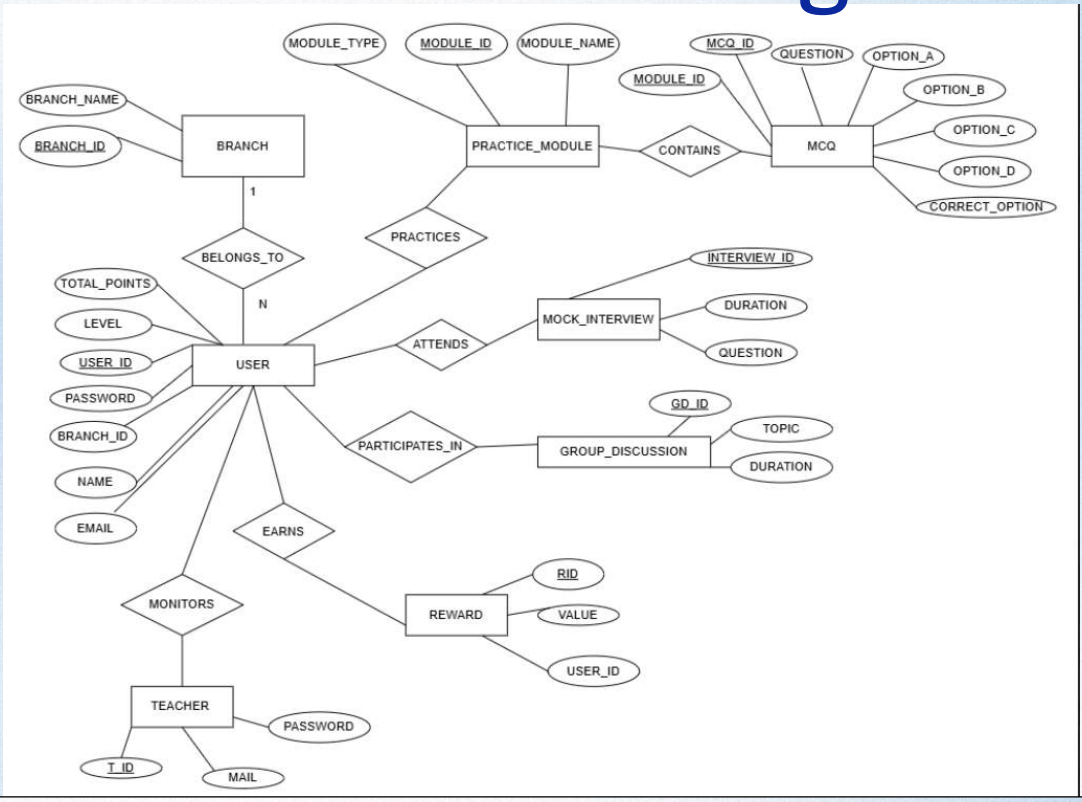
Adaptive Difficulty Adjustment Algorithm

1. Start
2. Initialize difficulty = Easy.
3. Conduct test and calculate score.
4. If score ≥ threshold: increase difficulty.
5. Else: keep difficulty same.
6. Gradually increase difficulty each week.
7. Update user profile and load next test.
8. Repeat.
9. Stop

Use Case Diagram

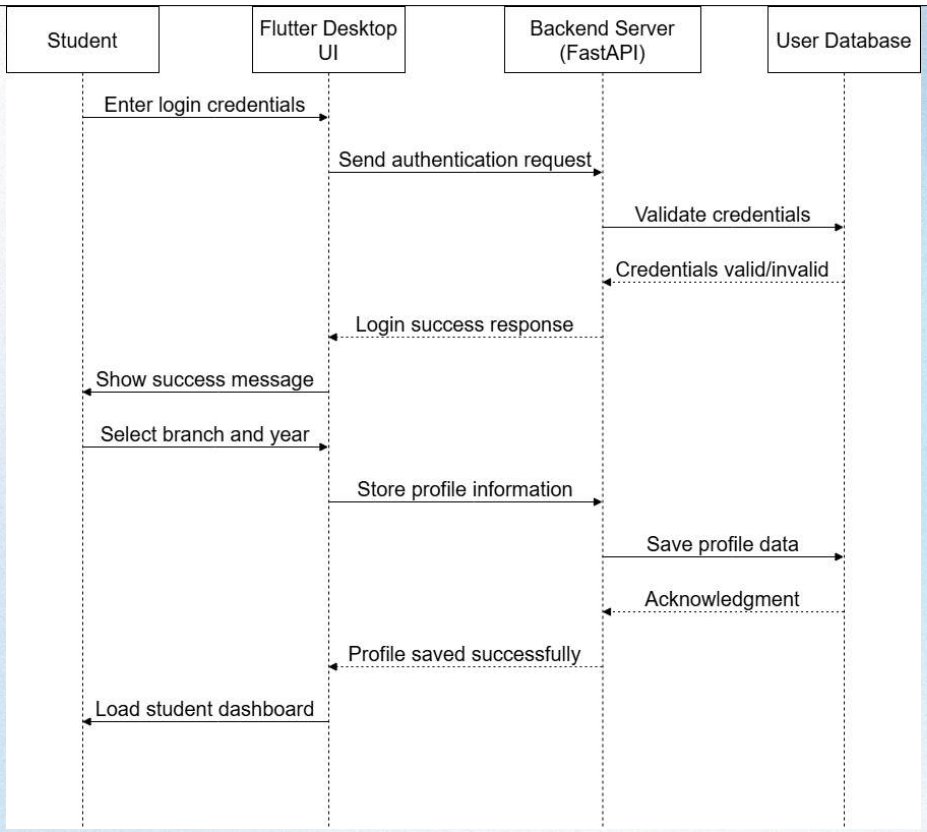


Database Design

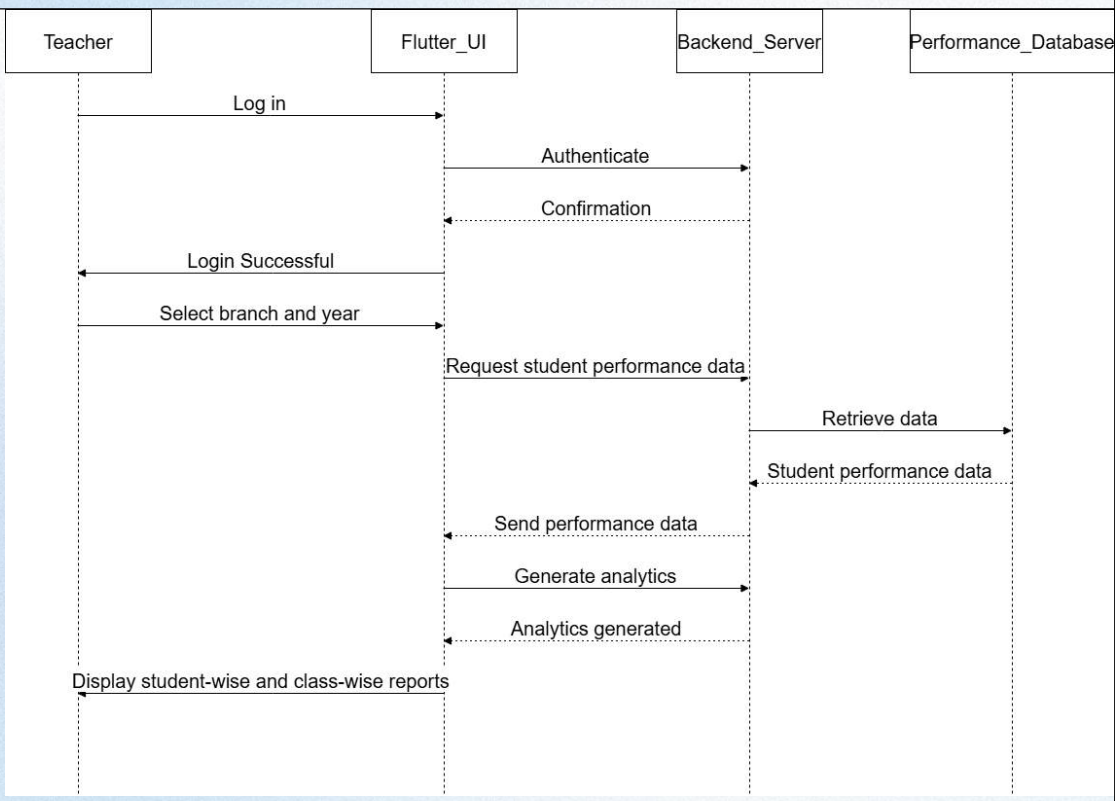


SEQUENCE DIAGRAMS

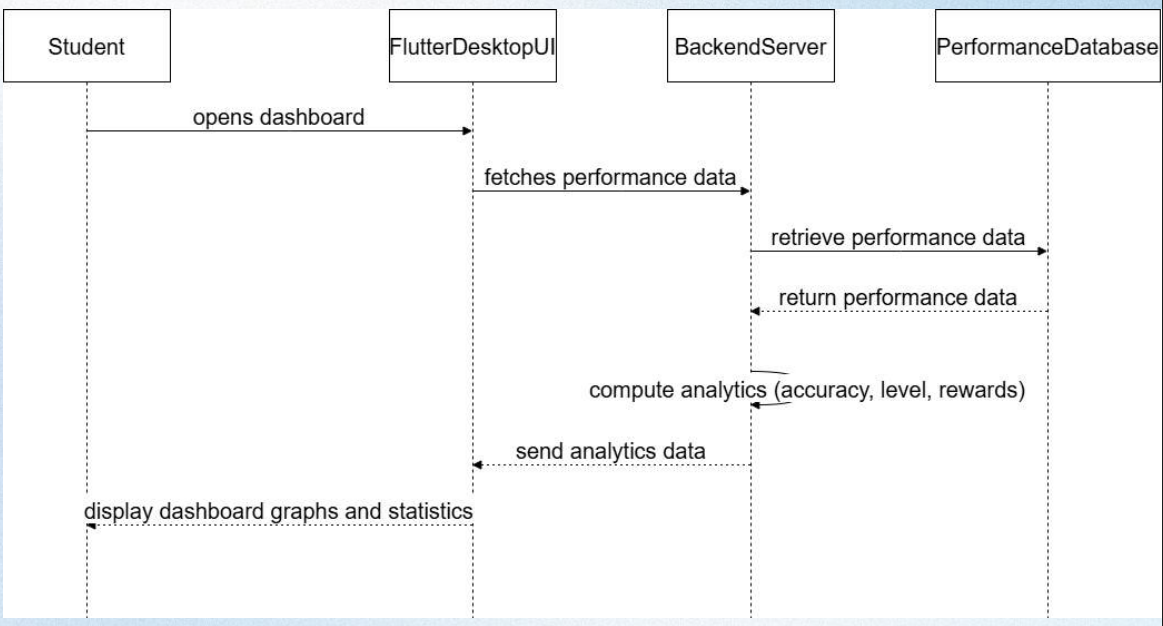
LOGIN



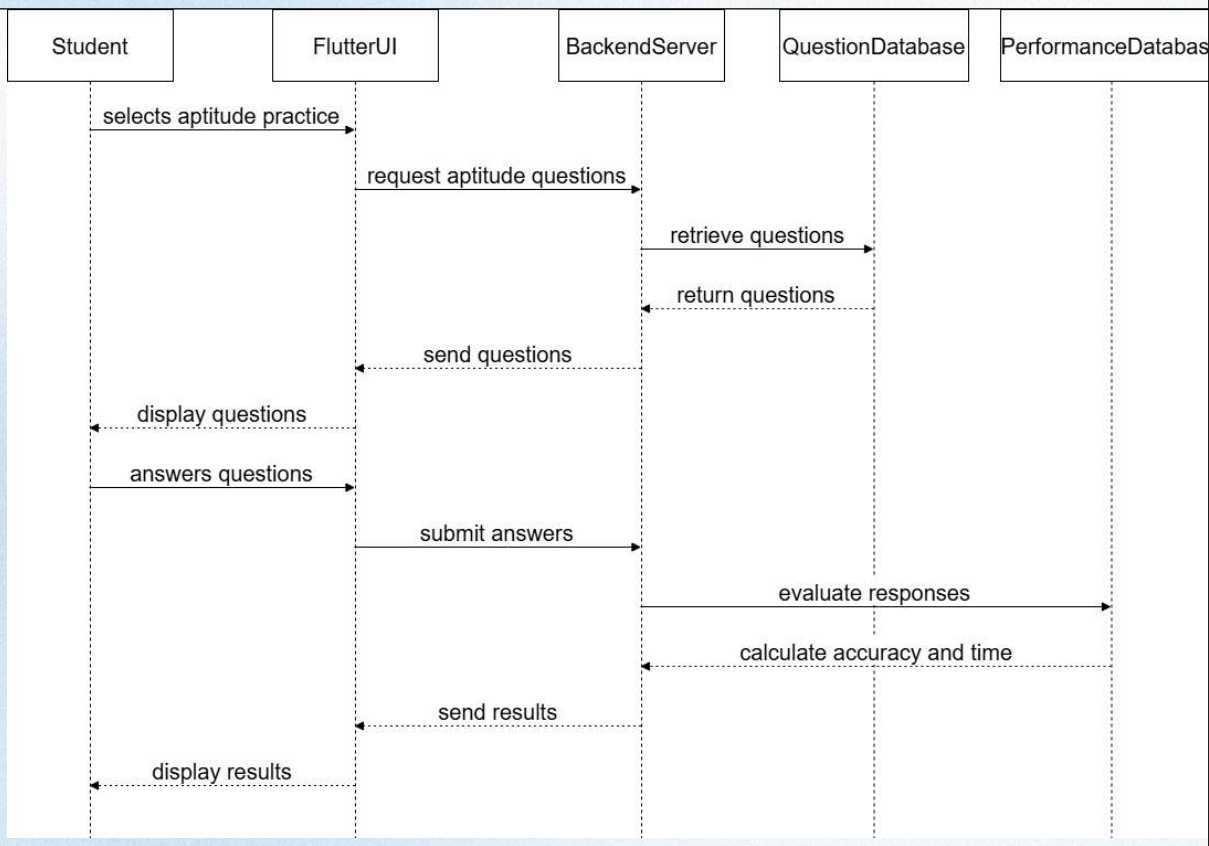
ADMIN (TEACHER)



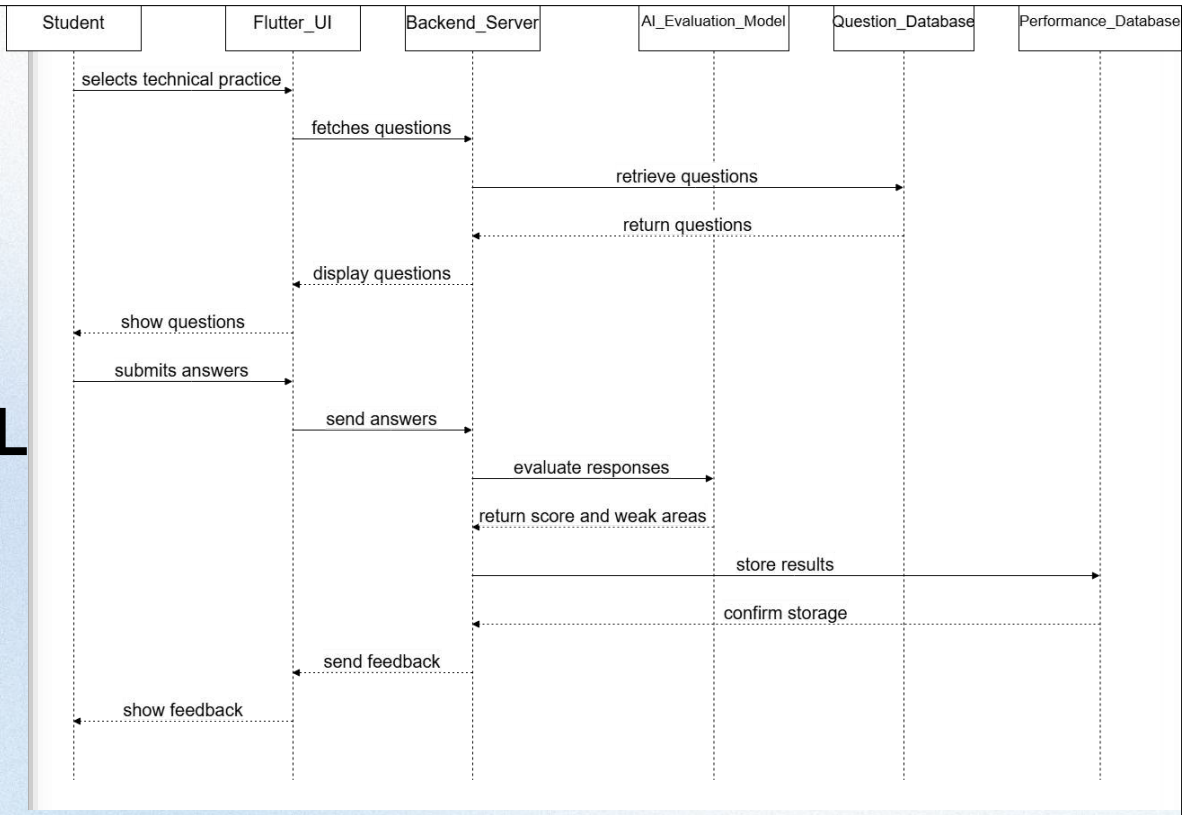
DASHBOARD



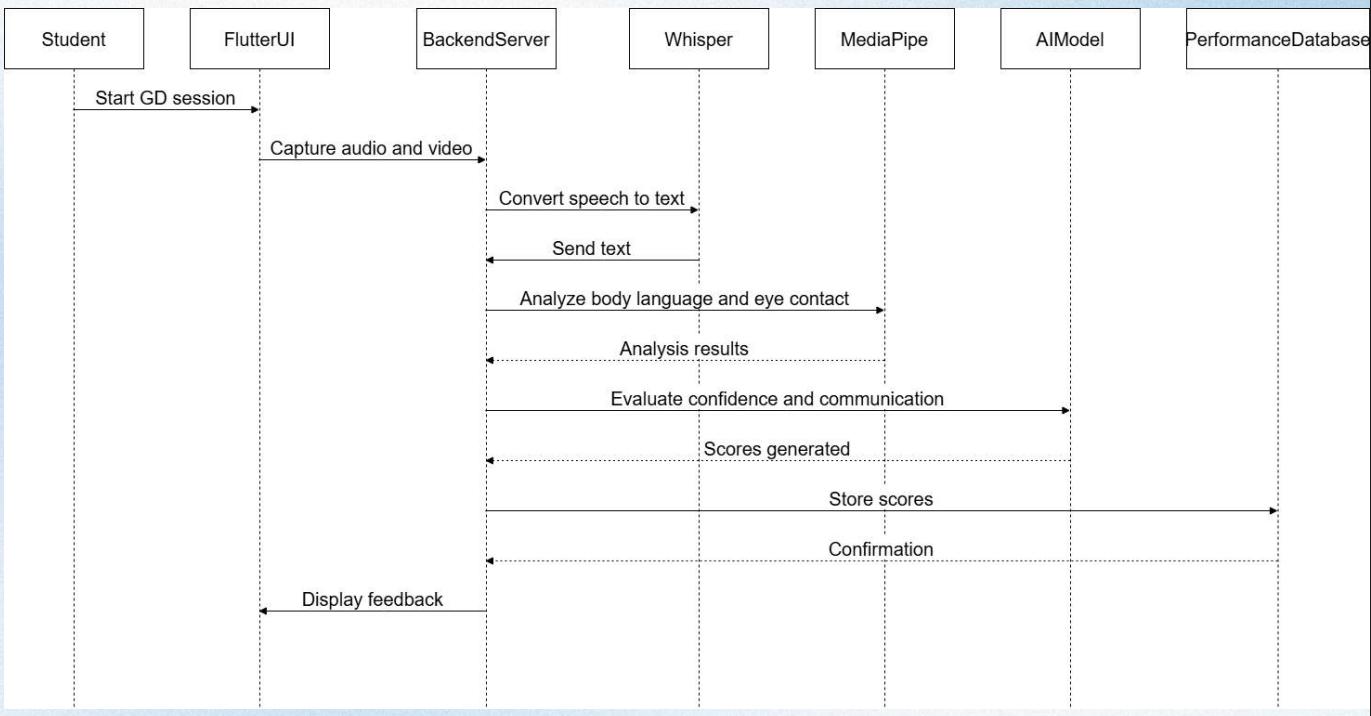
APTITUDE



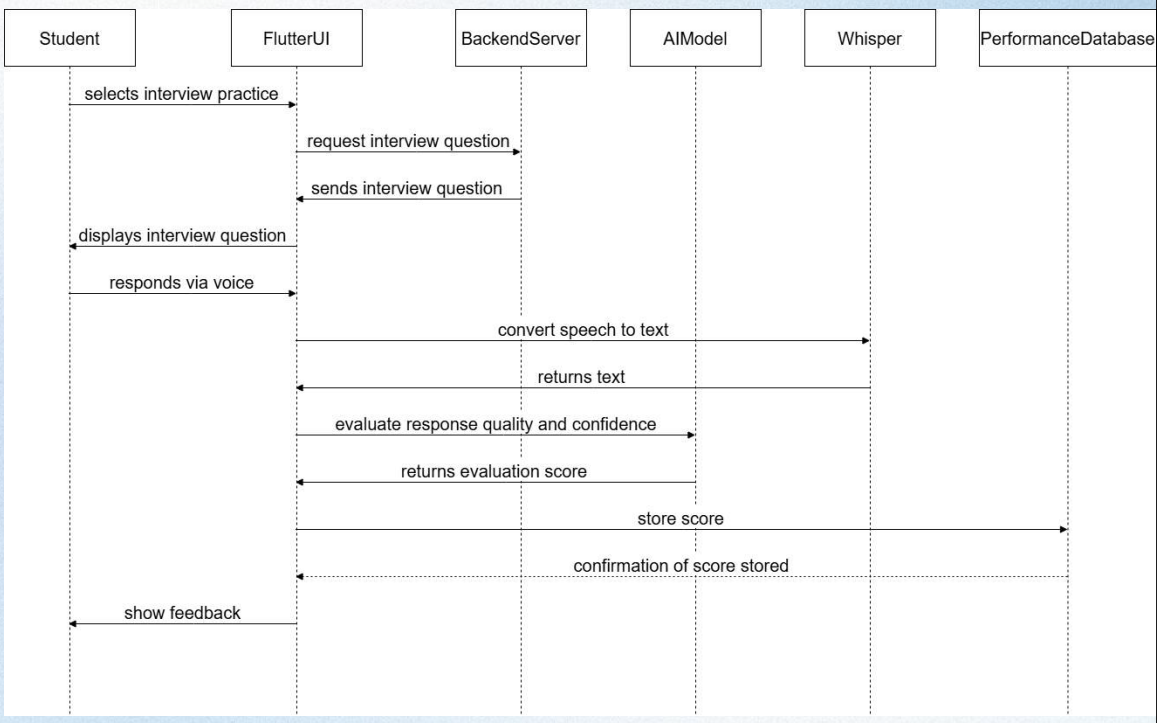
TECHNICAL



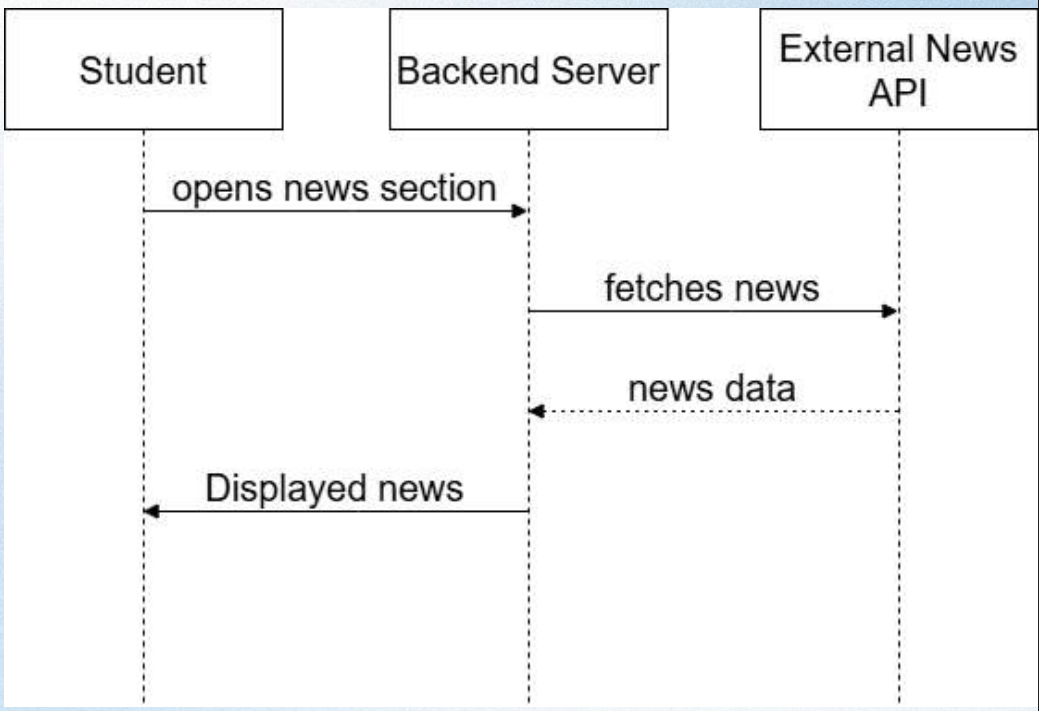
GD



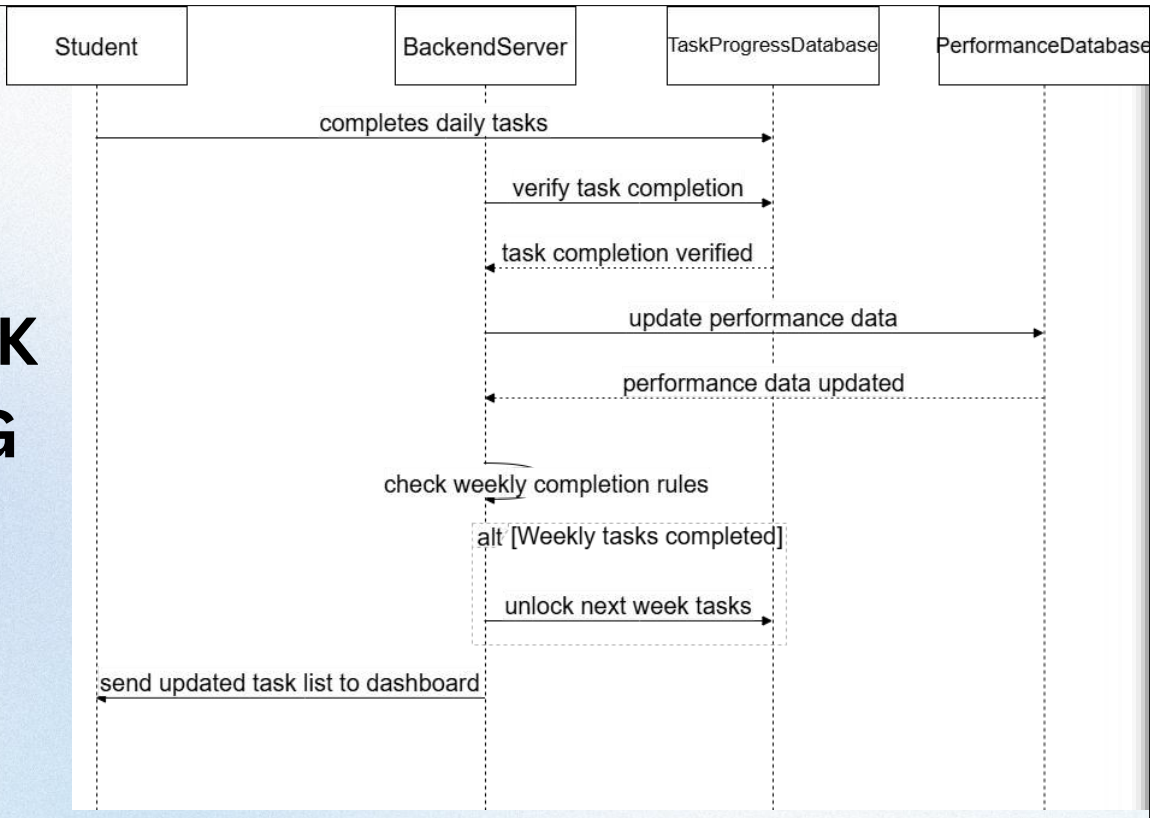
MOCK INTERVIEW



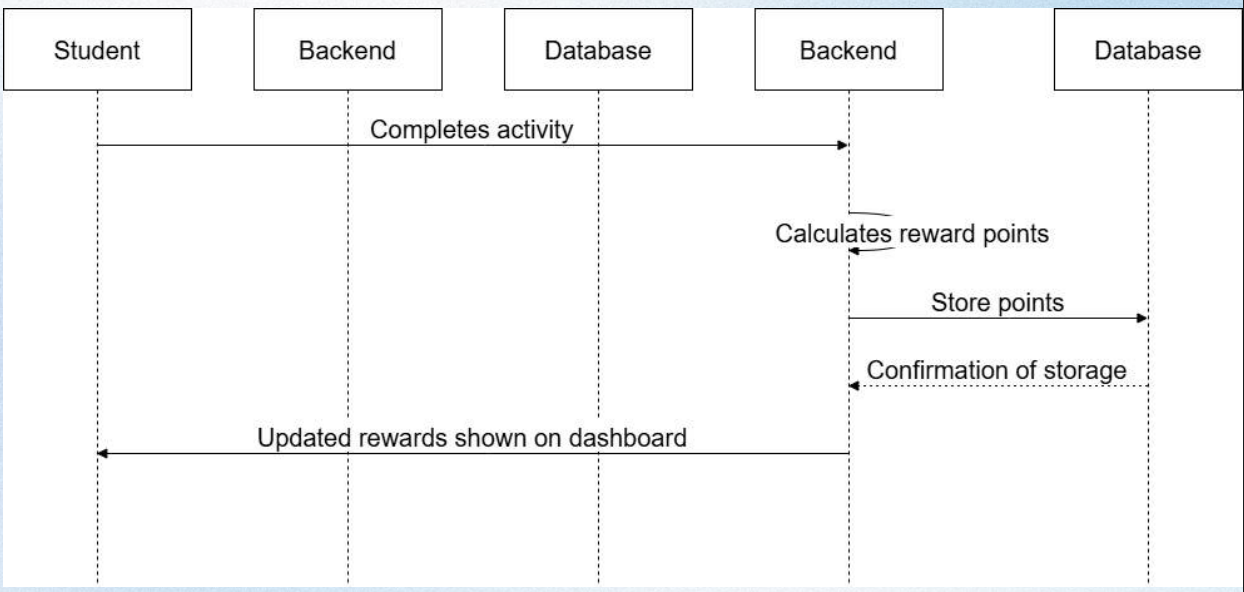
NEWS FETCHING MODULE



WEEKLY TASK UNLOCKING LOGIC



REWARD SYSTEM



Datasets

- Technical Question Dataset:

1. **CSE/IT/CSBS/AIDS** : https://www.kaggle.com/datasets/keithzidandsouza/engineering-aptitude-test-questions?select=cse_dataset.csv
2. **ECE/AEI** : https://engineeringinterviewquestions.com/electronics-communication-engineering-questions-answers/
3. **CIVIL** : https://engineeringinterviewquestions.com/civil-engineering-multiple-choice-interview-questions-and-answers/
4. **MECH** : https://engineeringinterviewquestions.com/mechanical-engineering-multiple-choice-questions-and-answers/
5. **EEE** : https://engineeringinterviewquestions.com/electrical-engineering-multiple-choice-interview-questions-and-answers/

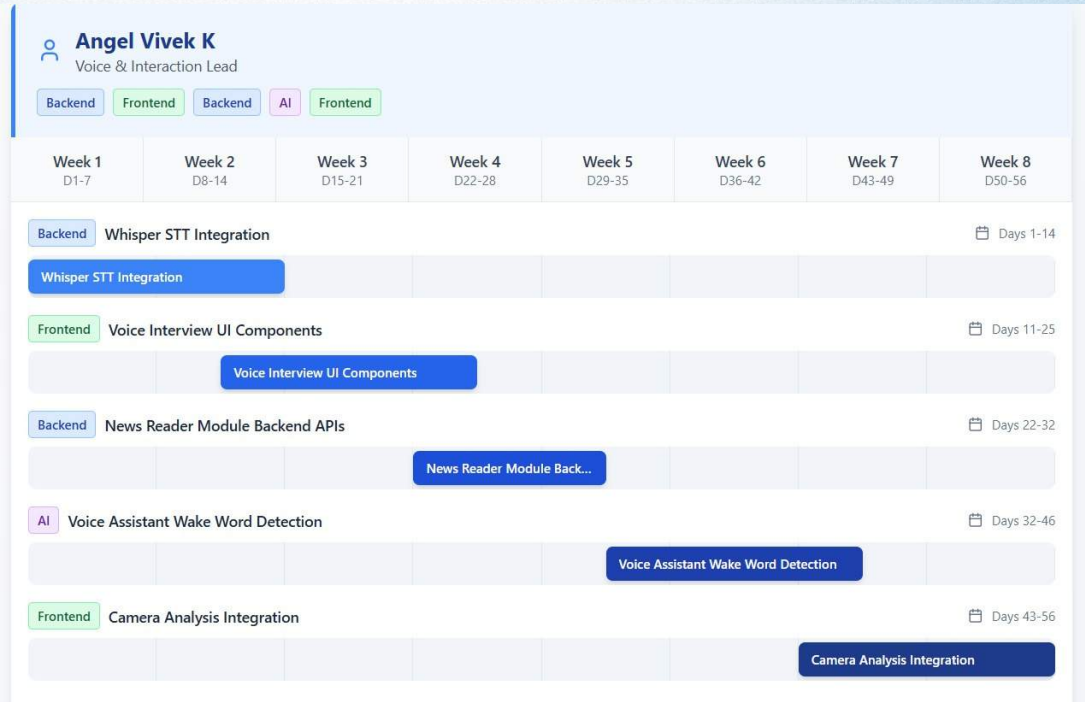
- Aptitude Dataset : https://www.kaggle.com/datasets/keithzidandsouza/engineering-aptitude-test-questions/code

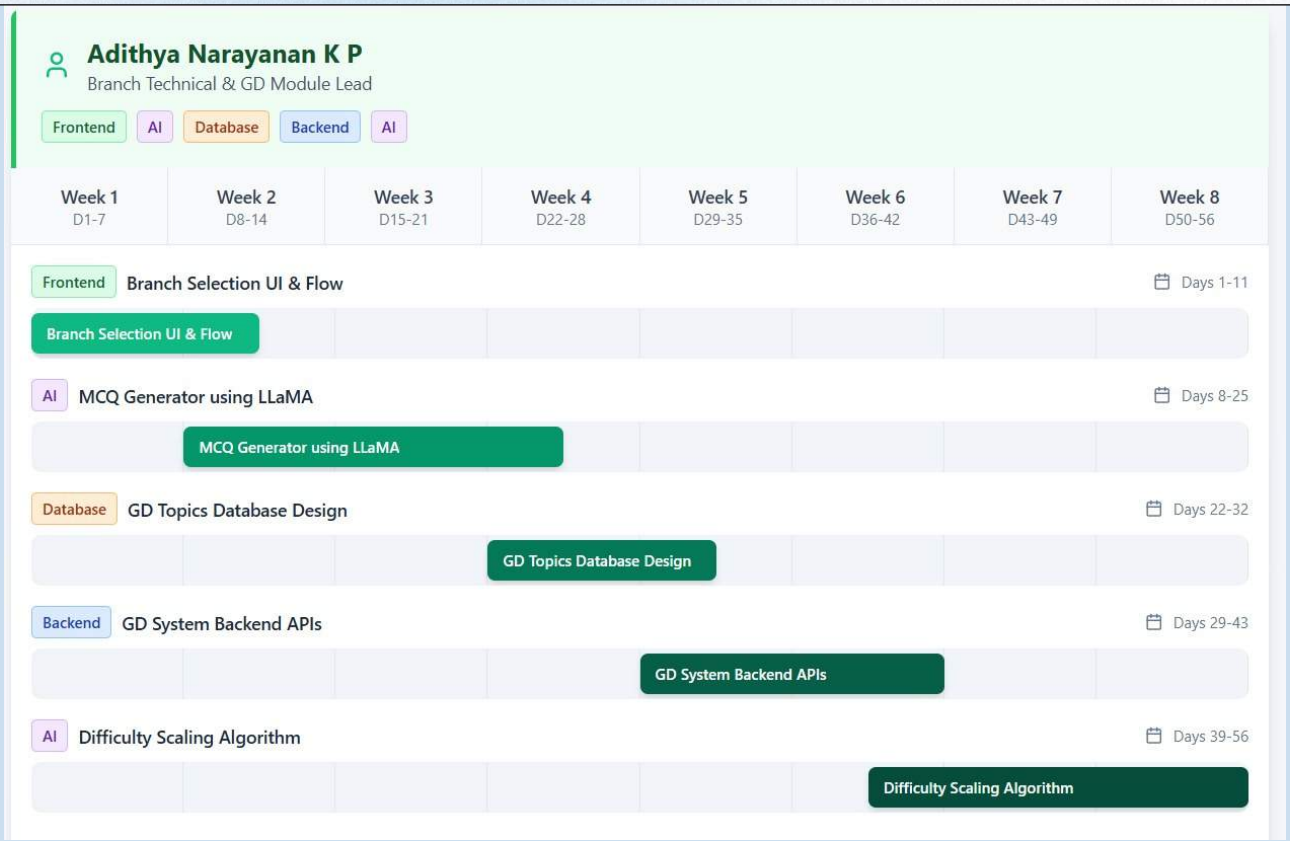
Datasets

- GD Topics Dataset: https://www.indiabix.com/group-discussion/topics-with-answers/
- Interview Topics Dataset:

1. **CSE/AIDS/IT** : https://engineeringinterviewquestions.com/computer-science-engineering-interview-questions-and-answers-pdf-cse/
2. **ECE/AEI** : https://engineeringinterviewquestions.com/electronics-communication-engineering-questions-answers/
3. **CIVIL** : https://engineeringinterviewquestions.com/civil-engineering-multiple-choice-interview-questions-and-answers/
4. **MECH** : https://engineeringinterviewquestions.com/mechanical-engineering-multiple-choice-questions-and-answers/
5. **EEE** : https://engineeringinterviewquestions.com/electrical-engineering-interview-questions-and-answers-eee/

Work Division





Anna Dantis

Backend & AI Evaluation Lead

Backend | Frontend | AI | Backend | AI

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
D1-7	D8-14	D15-21	D22-28	D29-35	D36-42	D43-49	D50-56

Backend — FastAPI Setup & Auth Backend — Days 1-14

FastAPI Setup & Auth Backend

Frontend — Interview Evaluation UI — Days 11-25

Interview Evaluation UI

AI — Answer Evaluation AI Engine (LLaMA) — Days 22-39

Answer Evaluation AI Engine (LLaMA)

Backend — Weekly Analytics Engine Backend — Days 36-50

Weekly Analytics Engine Backend

AI — Weak Area Detection — Days 46-56

Weak Area Detection

Avin Varghese

UI/UX & Database Lead

Database | Frontend | Backend | Frontend | Database

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
D1-7	D8-14	D15-21	D22-28	D29-35	D36-42	D43-49	D50-56

Database — Database Schema Design (SQLite/PostgreSQL) — Days 1-14

Database Schema Design (SQLite/Pos...

Frontend — Login, Signup & Dashboard UI — Days 11-25

Login, Signup & Dashboard UI

Backend — Gamification Backend (Points & Rewards) — Days 22-36

Gamification Backend (Points & Rewa...

Frontend — Analytics Dashboard UI — Days 32-46

Analytics Dashboard UI

Database — Database Optimization & Vector Storage — Days 43-56

Database Optimization & Vector Stora...

Software / Hardware Requirements

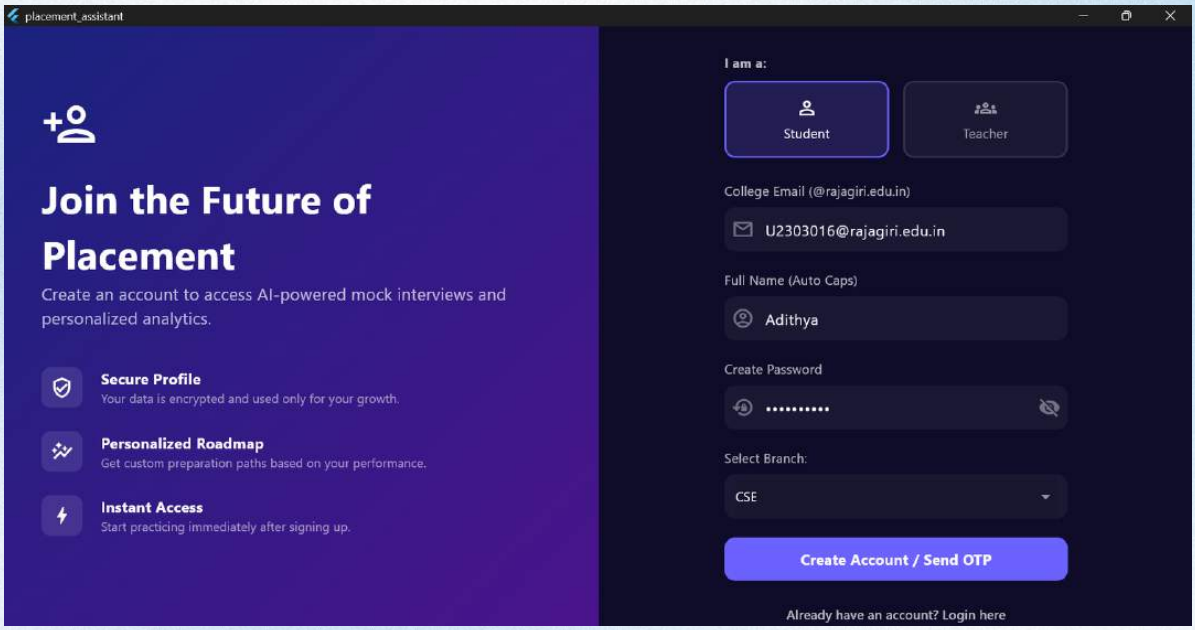
Software Requirements

- Flutter
- Python FastAPI
- Flutter Local Notifications
- REST APIs
- SQLite
- Porcupine
- LLaMA / Groq API
- Whisper
- (TTS)-Flutter TTS / Coqui TTS
- OpenCV / MediaPipe
- HackerNews API, RSS Feeds

Hardware Requirements

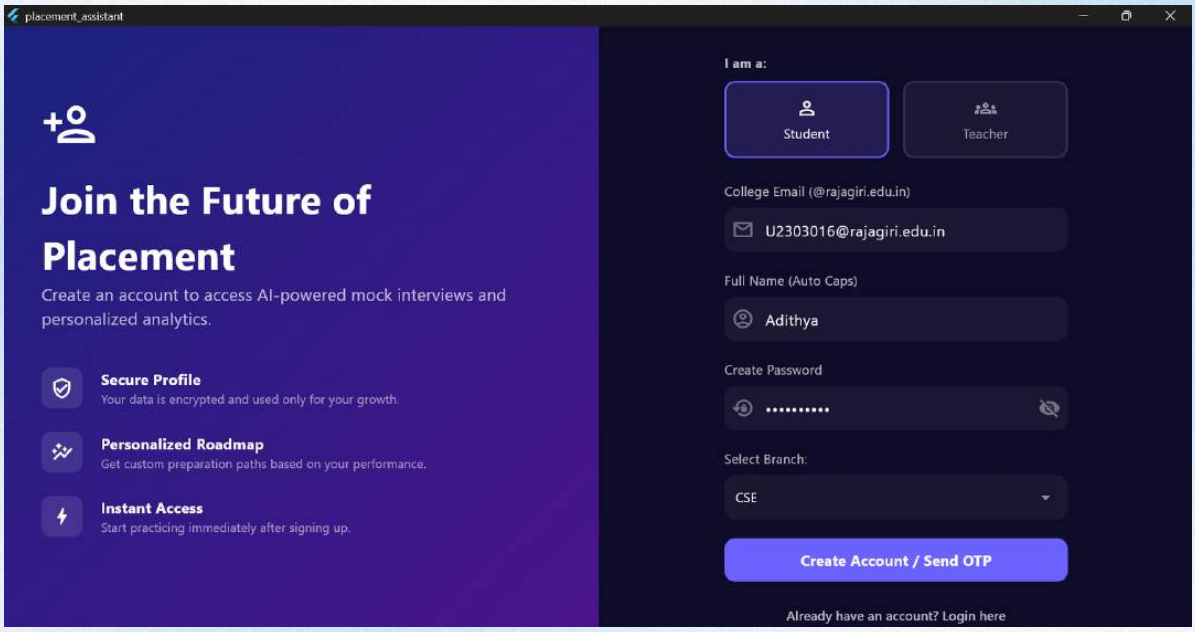
- PC / Laptop with minimum Intel i5 or equivalent processor
- Minimum 8 GB RAM
- Webcam for GD and interview practice
- Microphone / Headset for voice input
- Stable Internet connection

Results



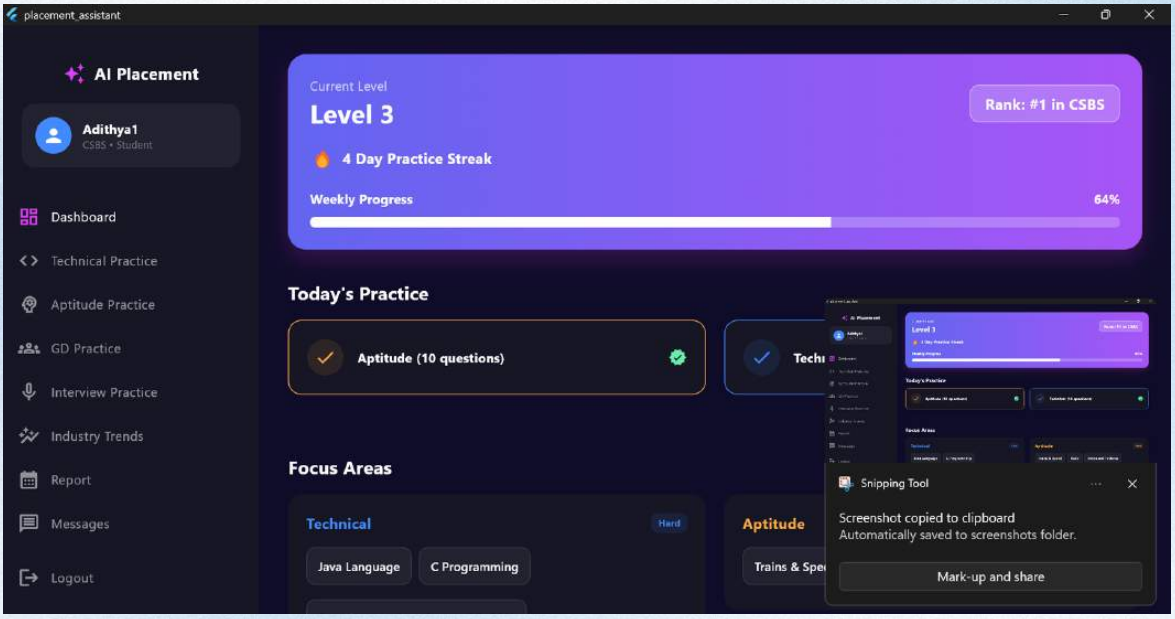
Student registration page

Results



Student registration page

Results

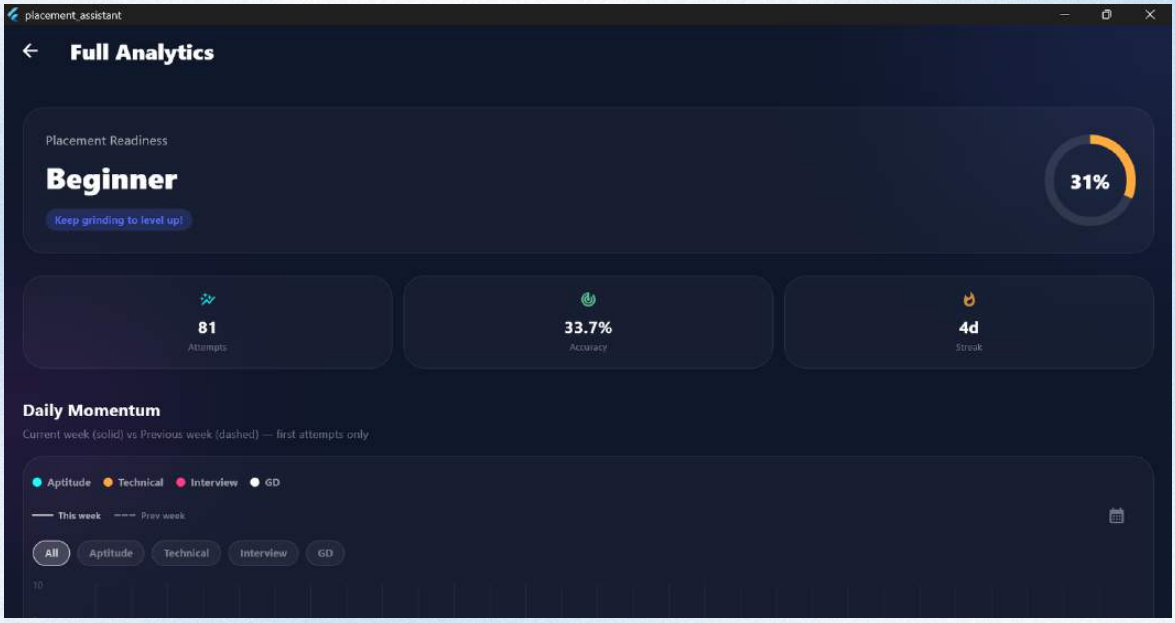


Student Dashboard

Results

Student Dashboard

Results



Student Dashboard-Full Full Analytics

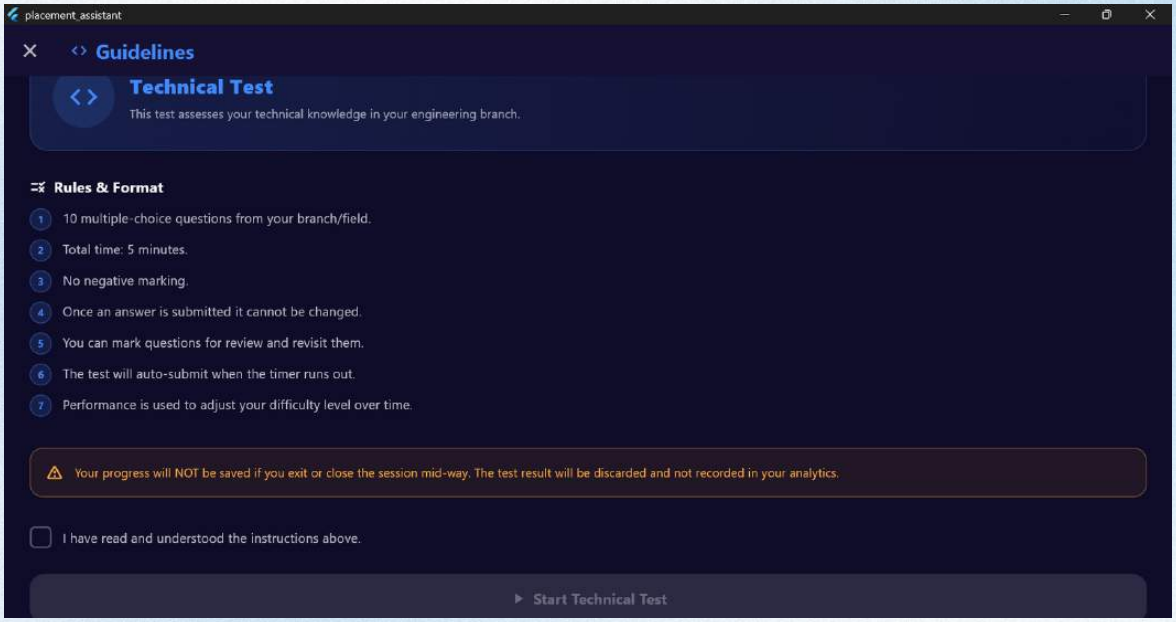
Results

Student Dashboard-Full Full Analytics

Results

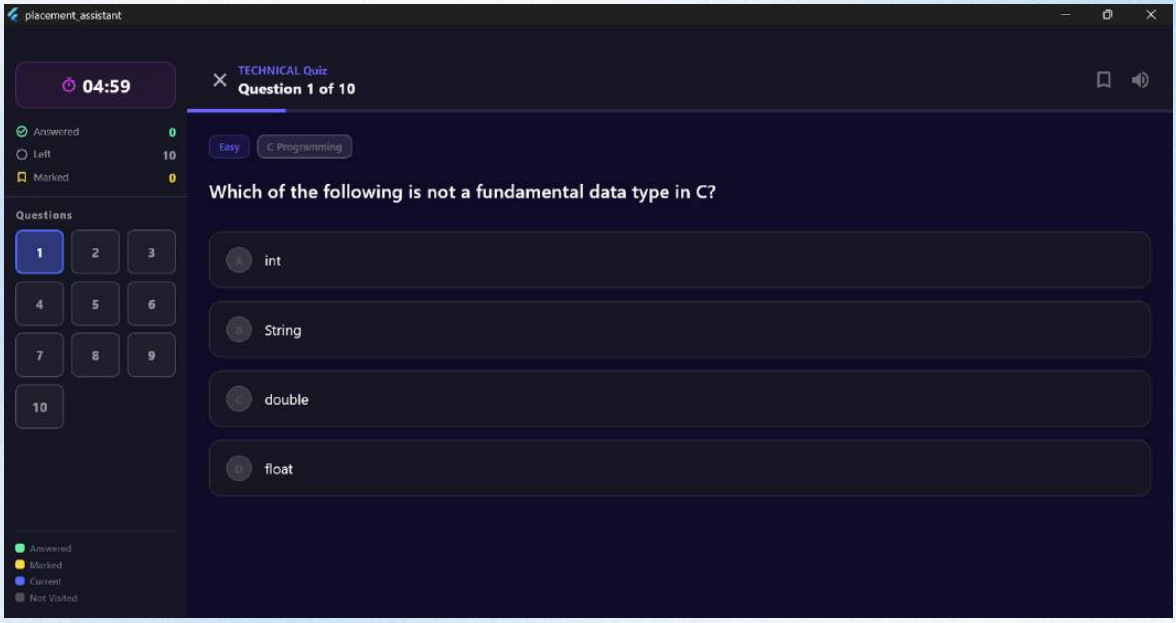
Previous work report

Results



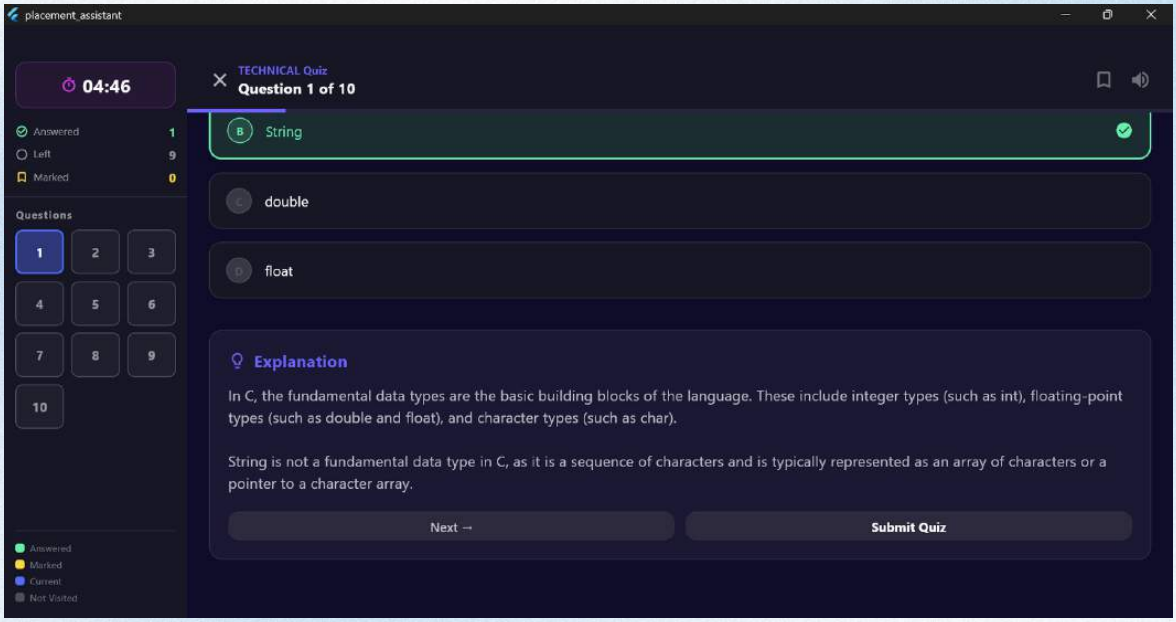
Technical quiz guidlines

Results



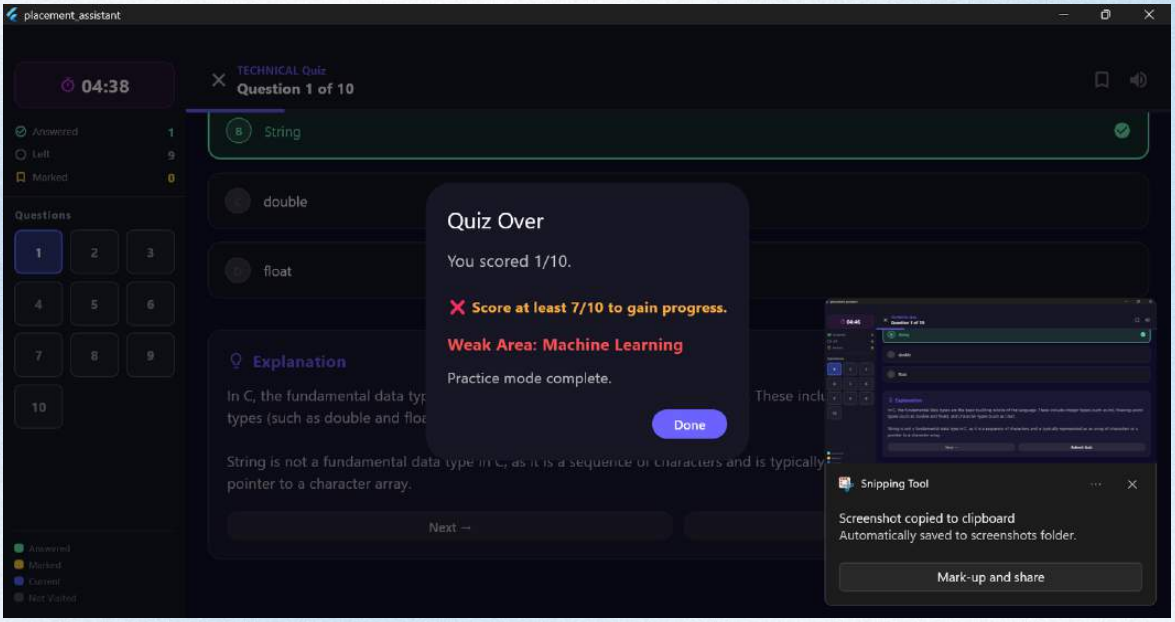
Technical quiz

Results



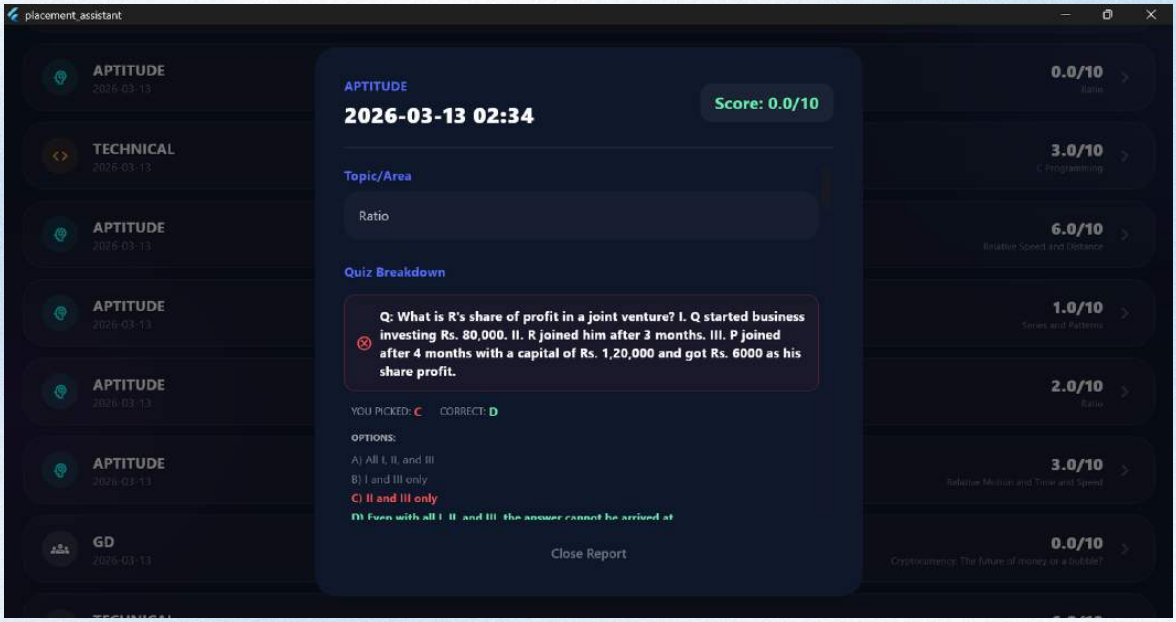
Explanation for answer

Results



After submit button - show mark

Results



Previous work report

Results

Previous work report

Conclusion

- The AI Powered Placement Assistant is a unified platform designed to enhance student's placement readiness.
- It offers branch specific technical and aptitude tests with adaptive difficulty and AI based evaluation.
- The system provides real time GD and interview feedback, weekly performance reports, and personalized recommendations.

Future Enhancements

1. Peer-to-Peer Group Discussion

   The system can allow multiple students to join real-time group discussions, record the conversation, and evaluate each participant's communication and participation.

2. Behavioral and Emotion Analysis

   The system can use the camera to analyze facial expressions, attention, and confidence during mock interviews.

3. Expanded Question Bank and Coding Practice

   More HR questions, soft skill questions, and coding problems can be added to help students prepare for different placement rounds.

References

- Shanmugam, P., Lenka, J., & Suhail, M. Y. (2024). Enhancing student placement preparation through web application. Available at SSRN 4834698.
- Nandunkar, K., Mhasde, C., & Shahabadkar, P. K. (2016). Enhancing placement opportunities for engineering graduates: A case study. In 45th ISTE National Convention and International Conference at Amravati [Access: 16.11. 2023]. Access on the internet: https://journaleet.in/articles/enhancing-employability-skills-and-placements-in-technical-institutes--a-case-study.

THANK YOU

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES (CO)

SNO	Description	Bloom's Taxonomy Level
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant applications.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND COURSE OUTCOMES (COs) – PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	Level	Justification
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.

Mapping	Level	Justification
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.

Mapping	Level	Justification
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.

Mapping	Level	Justification
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.